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Cascade utilization of low and medium enthalpy geothermal resources – A review

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ABSTRACT

This paper mainly presents a comprehensive review in different regions around the world about the use of geothermal resources of medium and low enthalpy in cascade manner. The utilization of geothermal resources in cascade levels is identified as a mean of sequential operation of geothermal heat by integrating different technologies for electricity generation, distribution and use of thermal energy, drying and dehydration processes, recreational uses, and any other direct-use of geothermal heat. Particularly, this review identifies cascade systems that have been implemented in a practical manner and conceptual systems that have been studied to determine their technical and economic feasibility. It is also observed the recent or in progress research that aims to demonstrate the advantage of using geothermal energy in cascade mode in order to maximize the use of geothermal resources in a rational and sustainable manner. The main features of the systems reported are highlighted, describing briefly the technologies used for these types of systems, temperature of geothermal resources, investment costs, methods of analysis and the main advantages and disadvantages. Finally, a discussion and some remarks are presented stating in general, that the use of geothermal energy in cascade is an effective and sustainable approach to promote the use of geothermal resources of medium and low enthalpy.

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1. Introduction

The consumption of primary energy in the world grows significantly every year. This is due to population growth, economic development and the coverage of the population's energy needs. This energy is based on a greater proportion on the use of fossil fuels with about 86%. Fossil fuels are being depleted and generate large amounts of CO_2, which contributes in greater extent to global warming effect [1]. This has caused the world to turn their eyes toward the search for different alternatives for the supply of reliable energy and decrease of environmental impact. Renewable energy resources are included among such alternatives.

Renewable energy resources, either for its present and future guaranteed availability, have significantly become a reality for better use of the natural resources and the reduction of pollution. The technological and sustainable development of this type of energy will help to solve the world's energy needs and challenges. In this regard, a renewable energy that stands out for its technological maturity is geothermal energy, which is considered renewable because of the enormous amount of energy available [2]. Geothermal energy comes from the natural heat of the Earth, which has an increase of internal temperature as depth increases away from the surface. However, the most important relationship between temperature and depth is given by geothermal gradient. In this sense, most geothermal exploration and use occur where the geothermal gradient is higher, [3]. Geothermal energy is one of the most promising energy alternatives and it is currently utilized in 24 countries worldwide to produce electricity and, in other 74 countries it is applied to various direct uses [4,5].

Geothermal resources can be classified as convective systems (hydrothermal), conductive systems and deep aquifers. In this regard, for hydrothermal systems and according to its temperature, it is possible to distinguish between high, medium and low enthalpy resources. It is known as a high enthalpy resource, when the geothermal fluid is under high pressure and temperature, usually above 150 °C. When the geothermal fluid is in temperatures between 100 °C and 150 °C, it is known as a medium enthalpy resource. It is a low enthalpy resource, when the temperature of the geothermal fluid is below 100 °C [2,6]. In this sense, geothermal resources of high enthalpy have been used to generate electricity in the world since 1904, but unfortunately the exploitation has focused almost entirely on these type of resources, and to a less proportion to the resources of medium and low enthalpy, mainly because of technological limitations. However, the use of medium enthalpy resource has a great potential that has been harnessed very little. It could be used to produce power by using technologies such as the Organic Rankine Cycle (ORC) and Kalina Cycle (KC) or through various direct applications [7]. In fact, a recent research indicates that for moderate temperatures of geothermal resources, ORCs integrated in hybrid renewable energy power systems are a most effective approach to reduce electricity generation cost, improve plant efficiency and extent the lifespan of reservoir, [8].

The direct utilization of geothermal energy is the oldest and most versatile way to harness geothermal energy of medium and low enthalpy. Current trending direct uses are mainly for heating systems working directly or through heat pumps, aquaculture, drying crops, growing plants and vegetables in greenhouses, processes of the paper and the cement industry, food processing, brewing, dyeing of fabrics, snow melting, cooling spaces and balneology, among others [9]. It is important to mention that in some cases the potential of medium and low enthalpy resources has not been used for power generation due to high initial cost and limited technology availability, such as in the case of China where despite of the high potential of geothermal resource, geothermal energy has been mainly employed for direct-use applications [10].

As a method to maximize the use of resources of medium and low enthalpy, both for electricity generation and for direct utilization, the use of geothermal energy in cascade has been proposed. This concept emerged in the mid 80s to make a more effective use of geothermal energy at different temperature levels. In general, the use of energy cascade is no more than a reasonable arrangement of the use of energy at different thermal levels to obtain different products [11–13]. Cascade utilization corresponds to a particular case of the integrated energy systems, which are also known as polygeneration systems, which focus on the principle of using one or more energy resources to provide various products more efficiently than the conventional systems, [11].

Based on the above, the main purpose of this paper is to present a comprehensive review of the integrated use of the geothermal resource in multiple applications under the concept called cascade utilization for power generation and sequential use

of geothermal heat for various direct uses, or by use of thermally activated technologies. Based on the review conducted, it will be highlighted that the concept of using cascading is an effective method to maximize the use of geothermal resources of medium and low enthalpy and it will also provide enough evidence to propose the implementation of these systems to other particular situations.

The review is organized as follows: Section 2 presents the concept of cascade utilization of geothermal energy, Section 3 describes the methodology followed for the preparation of this review. Section 4 presents the plants and systems in cascade documented in the literature around the world; it is organized by continents and their countries. Section 5 shows a summary of the cases reported and a table with the main features of each case. Finally, Section 6 presents a discussion and some concluding remarks.

2. The concept of the use of geothermal energy in cascade

In geothermal applications for the production of electricity, including direct uses, the geothermal fluid progressively yields its heat in the process and usually suffers a progressive decrease in temperature. This is not to say that once the process for which the fluid was intended is completed it will not present the potential for a further use. In the case of electricity generation, it is possible that the fluid temperature at the outlet of the generation process has sufficient thermodynamic quality and can be useful at a sequential process in a second or third level of temperature. Such processes or direct uses are for instance: heating systems, hot water supply, intermediate drying processes of food or wood, and other direct uses of geothermal heat as aquaculture and balneology, [14].

The concept of cascade utilization can be basically described as the harnessing of geothermal heat at different thermal levels in sequential processes. There are several possibilities to utilize geothermal energy in a cascade manner, but according to the cases contained within this review, it is possible to identify two main groups: cascade systems with electricity production and thermal uses, and cascade systems comprising only thermal uses. Fig. 1 is presented in order to describe better the concept of cascade utilization of geothermal resource. This particular system is a three-level cascade with electricity production and some thermal applications. In this example, the geothermal resource of medium enthalpy is utilized in the first level of the cascade for the production of electricity. Afterwards, the geothermal resource that comes out of this process feeds the second level of the cascade for freezing or cooling purposes using thermally activated technologies such as absorption or adsorption effect machines. After this second use, the fluid can be used further for other purposes with lower temperature requirements to form the third level of the cascade. Such purposes might be dehydration processes, greenhouses, balneology, etc.

The minimum average temperature requirement of most commercial low-grade heat converters, i.e. ORC modules, is 93 °C [15]. Therefore, in the case shown in Fig. 1, this temperature has been established as the minimum threshold to produce electricity in the first level of the geothermal cascade. The second level of the cascade is formed by considering also a temperature greater than 80 °C, which is the minimum temperature requirement of thermally activated technology (absorption or adsorption machines) for producing cold. Finally, the third level of the cascade is intended for direct uses that commonly demand a temperature lower than 70 °C. It is worth noting that temperature values shown in Fig. 1 represent values corresponding to this specific case formed by three cascade levels used solely for description purposes. However, the number of useful thermal levels is determined by the coupling between the temperature defined by the possible applications, and the temperature range between the initial temperature of the geothermal resource and a reference temperature that is usually above the environmental temperature.

3. Methodology

In order to accomplish the documentary review of integrated systems that use geothermal energy in cascade, a comprehensive search was performed in diverse information sources, such as databases of scientific papers, technical journals, conference presentations, technical reports, etc. In each source consulted, cascade utilization of geothermal energy is primarily highlighted by identifying the characteristics of the geothermal resource, the energy interactions of the systems and the technologies employed. The thermal levels of the cascade and the products obtained are also identified together with production capacity, characteristics of technologies and the capital costs of equipment or the facility in general. In particular, the review will mainly focus on the most relevant projects under development or that have been installed worldwide, where geothermal energy is used in cascade. However, this review also takes into consideration the projects and research that have not yet been implemented, but can contain relevant information to future proposals. The research and projects found are organized by continents (regions) and the countries belonging to that continent. This organization allows to display and to highlight the application and evolution of the concept of cascade utilization of geothermal energy oriented towards the use of geothermal resources of medium and low enthalpy. Finally, a table

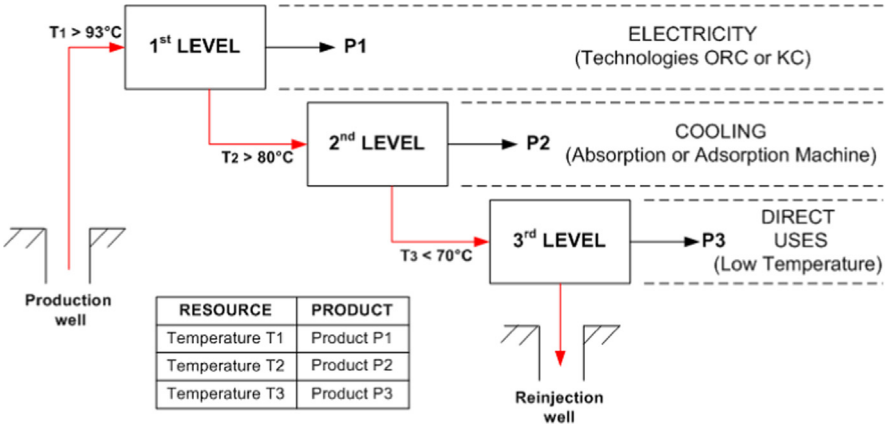


Fig. 1. Conceptual diagram of the cascade utilization of geothermal energy.

that summarizes the main characteristics of the systems found is presented, including thermal levels of the cascade and the products generated.

4. Review by region

4.1. Africa

This continent has only identified a case study of the cascade utilization of geothermal energy in Eburru community in Kenya. It may be noted that this is one of the African countries with the largest geothermal exploration and the first country to harness geothermal energy for power generation on the continent [16].

4.1.1. Kenya

The first documented case on the cascade utilization of geothermal energy arises in the community of Eburru, Kenya. The study proposes that the geothermal energy in two shallow wells is used for the drying of agricultural products, greenhouse heating, purification of honey and incubation of poultry products, as well as incorporating a recreational steam facility for a sauna and covering potable water needs. The cascade of this project is shown in Fig. 2 (Ref. [17]). In particular, the evaluation of the potential of the geothermal energy and the study of the chemistry of the vapor condensates of the wells were conducted. Assessing the technical feasibility was also performed with a proposal for drilling a new well for the project as additional energy supplement. The temperature of the geothermal resource in the wellhead is 89.6 °C and the applications of the cascade use temperatures above 50 °C. The cost of implementing this small project has been estimated at 40,000 USD, [17].

Another recent study is documented for the community of Barrier (Samburu District) where it is proposed a concept known as Geothermal Village for the development of rural areas. This concept takes into account an integral geothermal development to meet the demand for electric and thermal energy in human settlements located near the origin of the geothermal resource [18]. The Geothermal Village concept as an integral form of cascade utilization that includes electricity production by small ORC plants by the use of shallow geothermal wells, heat for drying food and use of geothermal fluids for sanitary applications. Other uses are considered such as ecotourism, pumping groundwater for cattle feeding and watering of small perimeters for crops and vegetable production (See Fig. 3, from Ref. [18]). For the integral system, it is considered a temperature of the geothermal resource of 120 °C with wells drilled to a depth of 500 m and a flow of 55.6 l/s with the capacity to produce between 1 MW and 2 MW of electricity.

4.1.2. Region summary

Cascade systems reported for Africa are located only in the geothermal zone of Kenya. Direct use applications are highlighted for shallow wells, focusing on drying of agricultural products and food. The temperature range suitable for cascading direct uses is 89 °C in wellhead and an outlet temperature for the last level of the cascade of 50 °C. However, the cascade concept can be extended comprising electricity production, when geothermal resource has 120 °C.

An important aspect to consider is that living standards of rural communities located near geothermal zones can be improved substantially by considering artisanal and industrial developments based on geothermal energy, including cascade utilization. This integral concept is known as Geothermal Village. On the other hand, climate conditions influence strongly some uses such as space heating, not reported for this continent.

4.2. America

In the Americas several countries have geothermal power plants and direct uses have also been reported, among the most important countries are Canada, Mexico and the United States. In regard to Canada, the proposal for an integrated system of electricity generation, heating, cooling and hot water production is observed. Wherein in Mexico, the conceptual proposal of cascade utilization of geothermal energy for an existing geothermal well in Maguarichi,

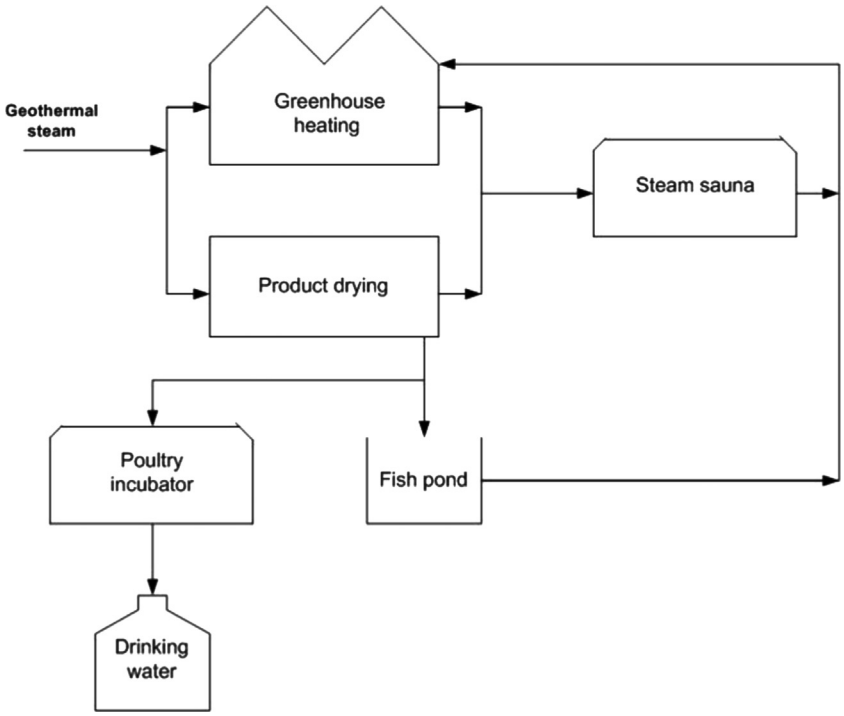


Fig. 2. Schematic of geothermal energy use in Eburru, sketched from [17].

Chihuahua is presented. For the United States of America, there are electricity generation plants with heat recovery for direct utilization in various applications. However, the most important cascade installation documented is Canby, California.

4.2.1. Canada

In Canada, researchers at the University of Ontario have proposed and analyzed an integrated generation system for electricity, heating, cooling and hot water production. The system consists of a multi-stage binary cycle with isobutane as working fluid, which together provide electricity, hot water, heating and cooling for a building. For the proposed system an evaluation of the thermodynamic performance is carried out, by evaluating the energy and exergy efficiencies and by identifying operating parameters for various temperatures of the geothermal fluid. Finally, a general evaluation of the system for practical applications is completed [19]. Fig. 4 (From Ref. [19]) shows the integrated system consisting of a Quadruple Effect Absorption System (QEAS) and a binary cycle plant that supplies power to a building. In this system the geothermal water is first passed through the cooling system. The cooling and heating effect produced are supplied to the building. Subsequently, the geothermal fluid passes the binary cycle power plant to produce electricity. A part of the energy produced by the binary plant is supplied to the refrigeration system itself and the remaining power is supplied to the building. The geothermal water from the binary cycle plant is supplied to the building for use as domestic hot water.

4.2.2. Mexico

In the case of Mexico, geothermal resources of medium and low temperature have been little used, solely for recreational purposes in thermal spas. The Federal Electricity Commission (CFE in Spanish) has been responsible for electricity generation, including geothermoelectric projects. Most of them have been successful, using geothermal wells with temperatures above 200 °C. Wells proven unsuitable for power generation had no further use or were abandoned. However, these wells reported temperatures between 90 and 150 °C that can be appropriate for power generation through binary cycles or for direct utilization. The literature reports that a geothermal resource of medium enthalpy was used in the community of Maguarichi (Chihuahua) to implement a project for electricity generation using an Ormat binary cycle plant of 300 kWe. The geothermal well used

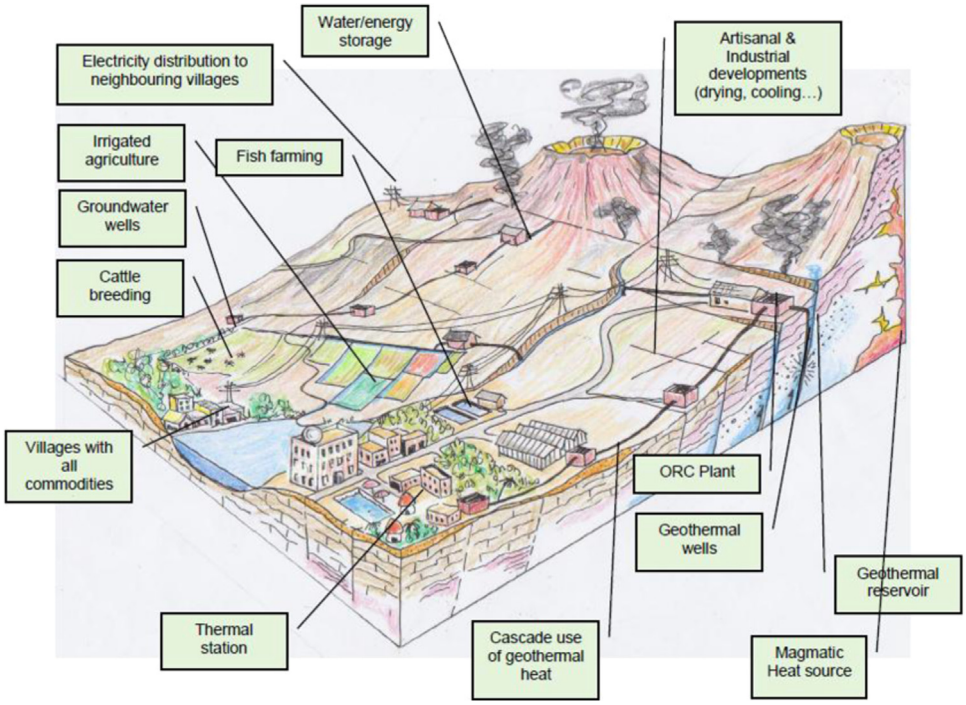


Fig. 3. Showing the conceptual diagram of the Geothermal Village for being developed in Samburu, Kenya, as taken from [18].

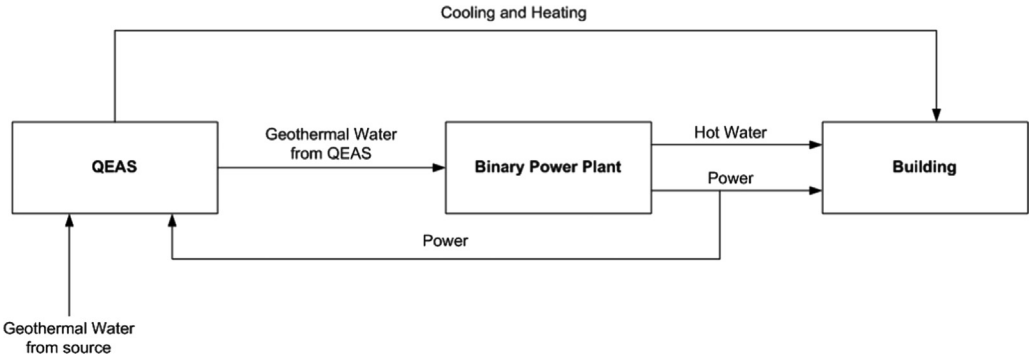


Fig. 4. Schematic diagram of the multi-product geothermal cascade system, sketched from [19].

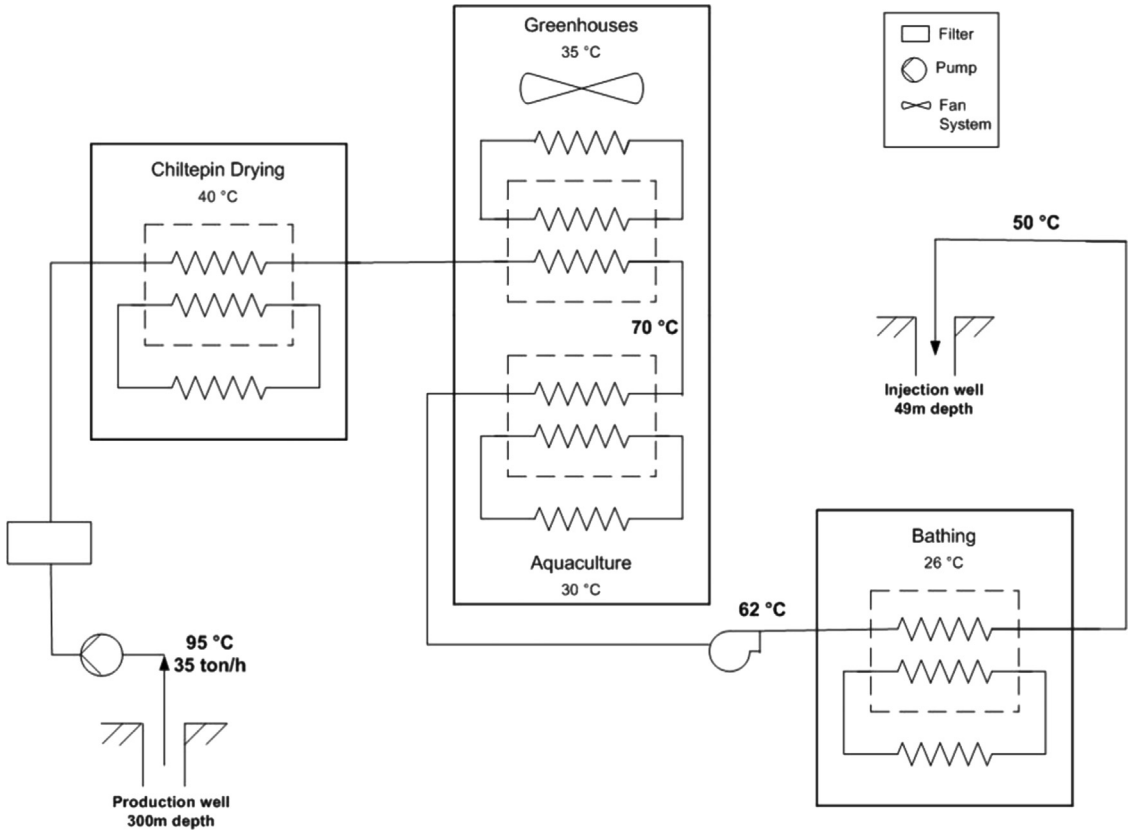


Fig. 5. Conceptual diagram of the geothermal utilization in cascade for the Maguarichi case, sketched from [20].

brings a mass flow of 9.72 l/s (35 t/h) and a temperature of 98 °C at a depth of 300 m. The plant stopped operating when the national electric grid reach the community, and later, it was donated to The Universidad Michoacana de San Nicolas de Hidalgo (UMSNH), [20]. In order to continue using geothermal energy in this community, a cascade system has been considered. Fig. 5 (Ref. [20]) shows the conceptual diagram of the proposal with four sequential direct uses, the first for drying a type of chili pepper called chiltepin, the second use is intended for greenhouses and aquaculture. Finally a use in balneology, with the subsequent reinjection.

4.2.3. United States of America

For the Institute of Technology in Oregon an application of the cascade utilization of geothermal energy that involves harnessing energy through a binary cycle and a subsequent use of the geothermal heat for fish hatcheries is documented [21] (see Fig. 6). The scheme considers using the effluent of the binary cycle plant with a temperature of 96.1 °C by decreasing it to about 24 °C, using a flow of 3.5 l/s. In this case, there is no other relevant technical data reported.

Another system with power generation and use of thermal energy is reported at San Emidio geothermal area, to the north of Reno, in Nevada. The installation consists of a power plant altogether with a plant for the dehydration of onion and garlic. The plant was built by the Ormat Company and was put into operation in the mid-1988. The plant consists of four units of 1.2 MWe, to jointly produce 3.6 MWe of net power at a design temperature of 141 °C (285 °F). The dehydration plant uses approximately the same temperature of geothermal water for the process of drying. Currently, the capacity of the dehydration plant is 34,019 kg (75,000 pounds) of onions per day or 38,555 kg (85.000 lbs) of garlic [22].

Another advantage with the formal designation of cascade utilization is reported for the region of Klamath Falls, Oregon, [23]. In the region there are 6 geothermal wells with temperatures ranging from 27 °C to 93 °C and depths around 90 m. The system makes use of cascade for agriculture and aquaculture. The main well is used for greenhouse heating at 93 °C. Subsequently, the greenhouses' water effluent at a lower temperature is used to heat 37 shallow ponds of tropical fish. The flow at the exit of the ponds is cooled in a large pond and is eventually used for irrigation.

A similar application to that found in Oregon, is in Cotton City, New Mexico. For this area it is considered installing a Kalina cycle plant with a net power of 1 MWe, using a mixture of ammonia-water as working fluid. The geothermal resource comes from a well with a depth of 120 m, with a temperature between 115 °C and 120 °C, and a flow of 63 l/s. This plant will supply electricity to a facility for fish hatchery and the waste heat will be also used to maintain the temperature of the hatchery. It is estimated that the cost of the system is of 3.3 million (USD) [24,25].

Between 2003 and 2005 in Chena (northeast of Fairbanks, Alaska), the first Ice Museum of the United States was built. For this region, the geothermal resources available are thermal springs and geothermal wells with an average temperature of 74 °C (165 °F) and 300 m of depth. To produce the ice in the museum, it was decided to use this geothermal resource by installing an absorption machine. The installed machine is of medium effect and has a cooling capacity of 53 kW (15 TR), which keeps the museum frozen; Energy Concepts has been the manufacturer. The absorption system uses three levels of pressure with the working pair (ammonia – water) reaching a temperature in evaporator of −29 °C and maintains the hotel with a temperature of −4 °C. The cooling water from the absorber and the condenser is taken from a

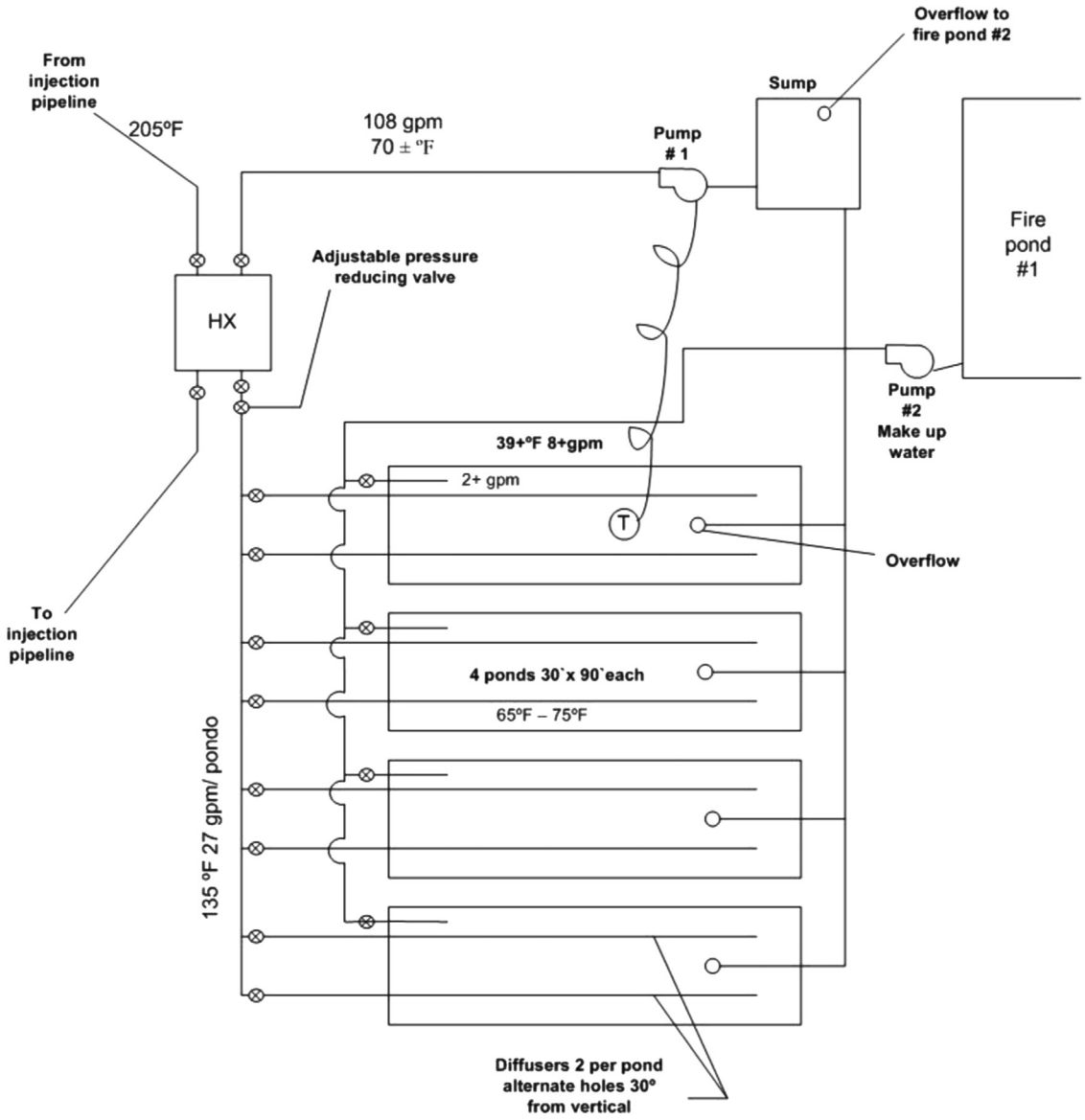


Fig. 6. Arrangement of the heat recovery, sketched from [21].

nearby river with a temperature of 4 °C, which improves the system performance. After passing through the cold generation process, the geothermal fluid is re-used applying the cascade concept to complete the heat requirements of a pool. It has been considered that the geothermal fluid from this process is later used for heating greenhouses, [26–28].

Finally for The United States, the most recent installation for the cascade utilization of geothermal energy is being developed in Canby, California, and it is expected to be completed by 2015. The facility consists of an ORC power plant of 50 kWe, greenhouses, district heating, and aquaculture [29,30]. The objective of this project is to produce electricity from the utilization of the geothermal resource with a temperature of 96.1 °C (205 °F) in addition of using the residual energy of the power plant to implement the cascade utilization in various applications of direct utilization. Fig. 7 shows the flowchart of the Canby project [31]. It can be seen that the geothermal resource enters the power plant first and the heat dissipated in the condenser can be used for aquaculture. Subsequently, upon leaving the evaporator of the power cycle, the geothermal resource is primarily used for district heating and then for heating greenhouses. Finally, the geothermal resource is reinjected.

4.2.4. Region summary

The Americas has an important potential of geothermal energy, but regarding geothermal utilization in cascade manner only three countries reported this type of sustainable energy use: Canada, Mexico and United States of America. Most of the systems reported have included electricity production in the first level of the cascade, with exception of Mexico and Klamath Falls (Oregon, USA) that reported that shallow wells with temperature less than 98 °C are appropriate for agriculture and aquaculture direct uses in different thermal levels. However, if temperatures range between 90 °C and 150 °C binary cycles or organic Rankine Cycles can be integrated increasing cascade levels. In this sense, old wells or exploration wells already drilled that were not suitable for power generation can be put into operation reducing capital cost.

In some cases, the final uses or products in the cascade system are conditioned by the climate zones favorably. For instance, with low enthalpy resource of 74 °C for Chena hot springs, it was possible to produce ice in the first level of the cascade by using absorption machines and part of the heat rejected was used as the second level for greenhouses. The overall performance of this system was increased substantially because of the cooling water

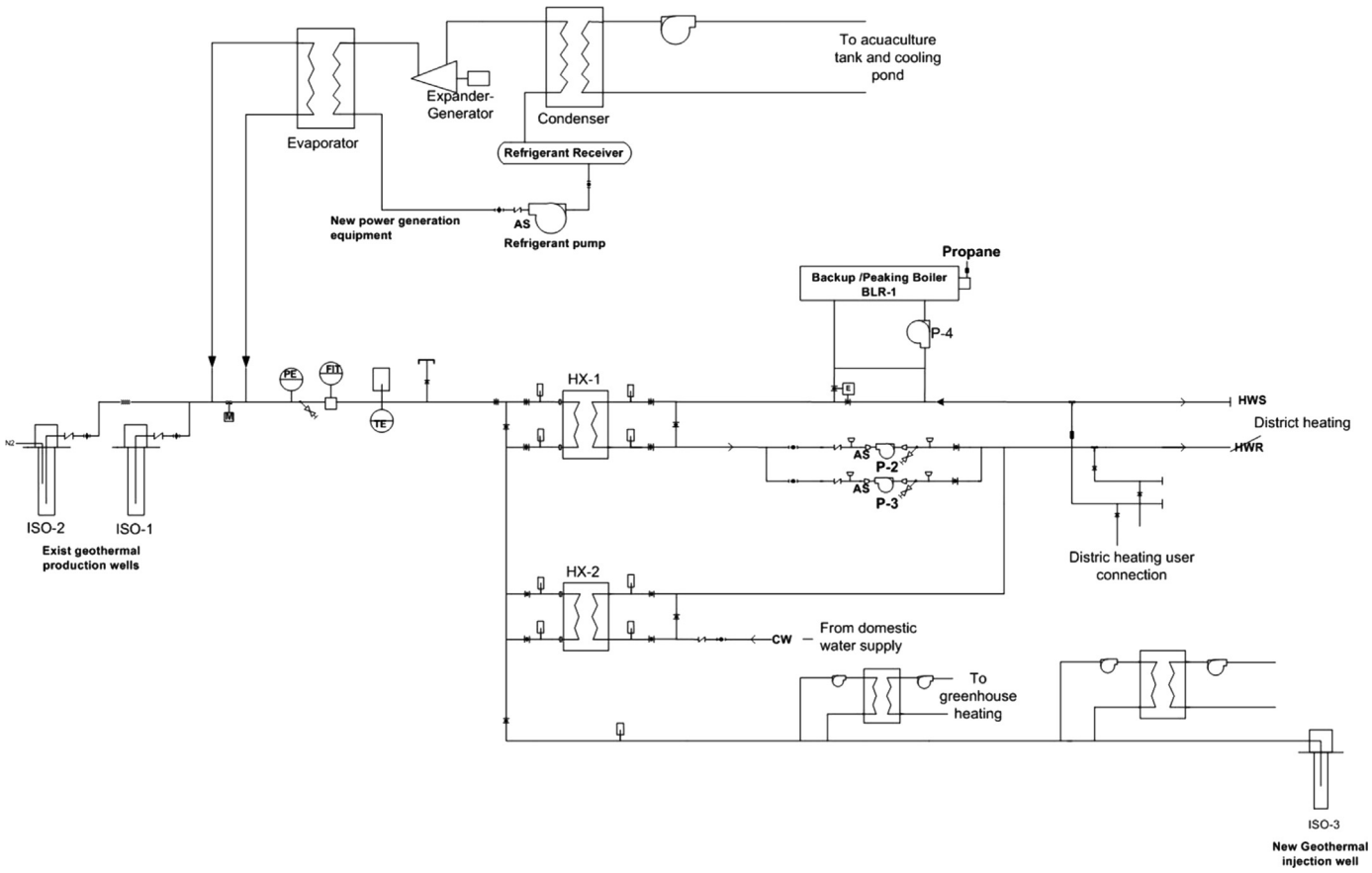


Fig. 7. Schematic Project System Canby California, outlined from [31].

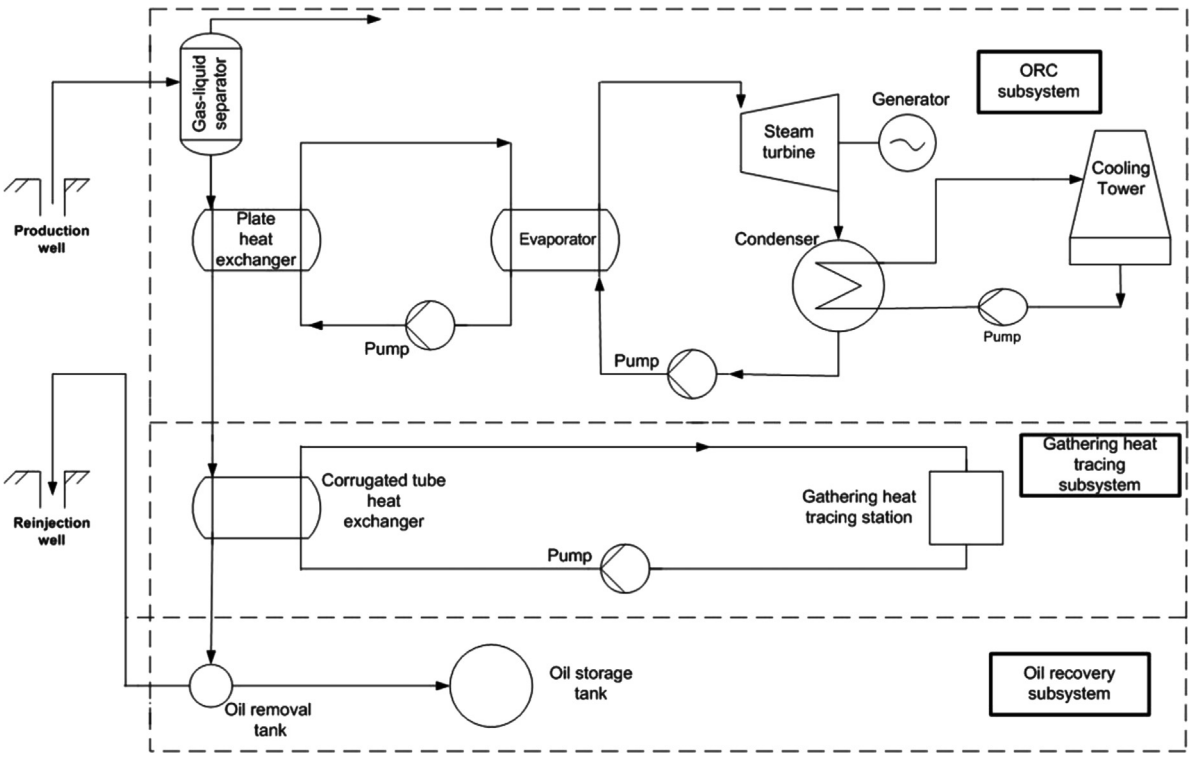


Fig. 8. Cascade utilization of geothermal energy in electricity production and an oil field, sketched from [37].

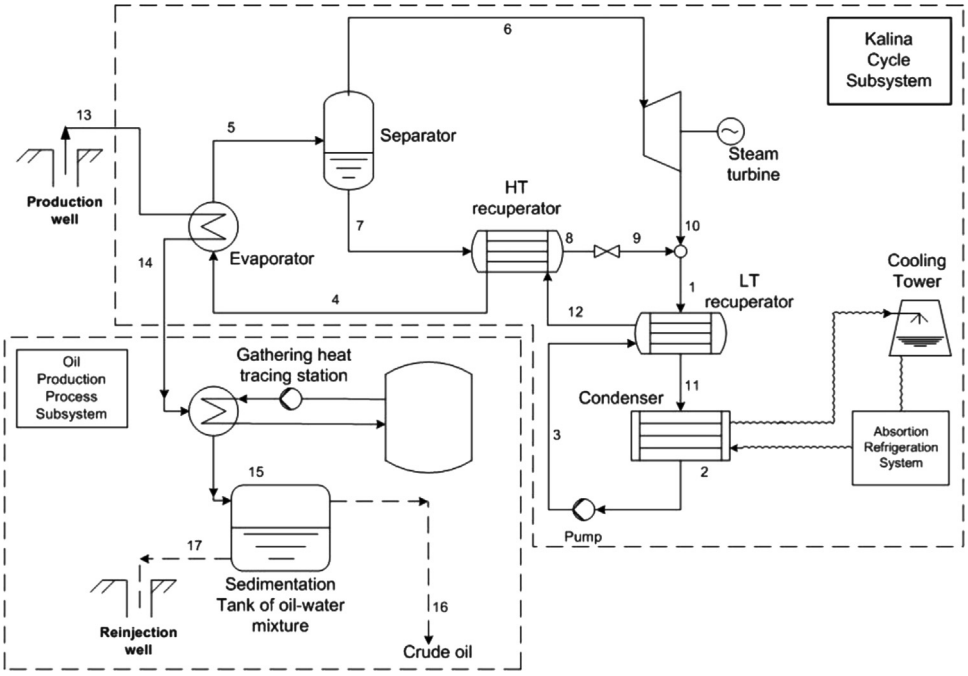


Fig. 9. Cascade system with Kalina cycle and production of crude oil, sketched from [38].

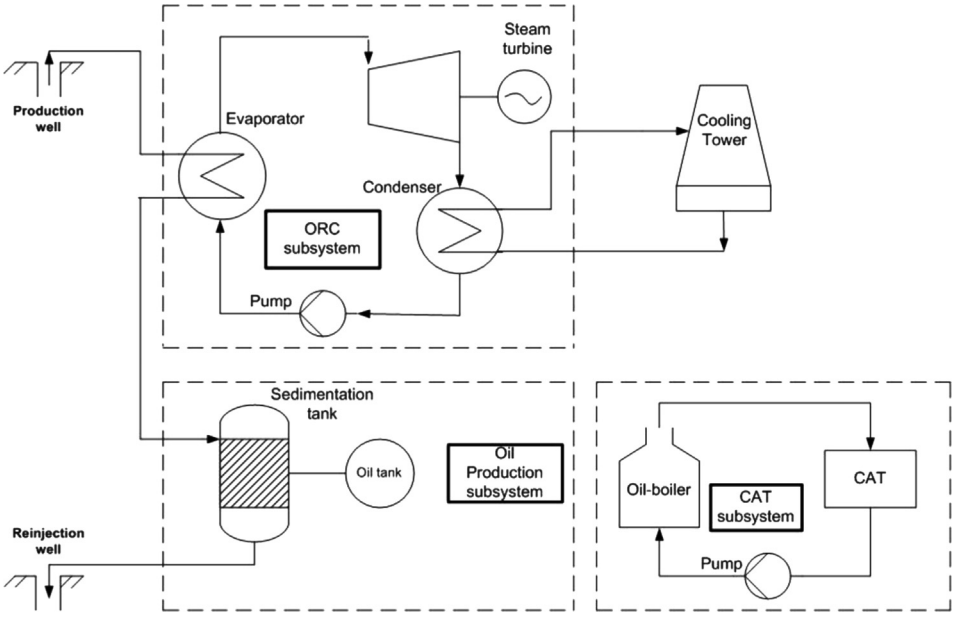


Fig. 10. Diagram of the cascade system based on ORC, outlined from [39].

for the absorber and condenser of the absorption machine was taken from a river whose temperature remains almost constant with 4 °C.

4.3. Asia

The Asian continent has many power plants operated by geothermal energy. In regard to multiple applications of geothermal energy, China, Iran and Thailand are the leading. In China the feasibility of implementing new geothermal technologies interacting with oil fields and processes is analyzed. As related to Iran, the direct utilization of waste heat from a power plant is proposed. In Thailand, installing a binary cycle plant of 300 kWe using the waste heat for drying processes and recreational uses is reported.

4.3.1. China

For China several publications are reported in which cascade utilization is considered as an effective, rational and sustainable way to use the geothermal resources of the country. Different perspectives on the integral cascade utilization of geothermal water are analyzed, applying technologies such as radiant heating, heat pumps for heating systems, plate heat exchangers, etc. Various techniques are applied to maximize the use of geothermal energy and to reduce the depletion of geothermal reservoirs. Studies show the potential of cascade utilization, plus getting significant social, economic and environmental benefits. The regions reported in this work are Tianjin, Qinhuangdao and Hengshui [32–36].

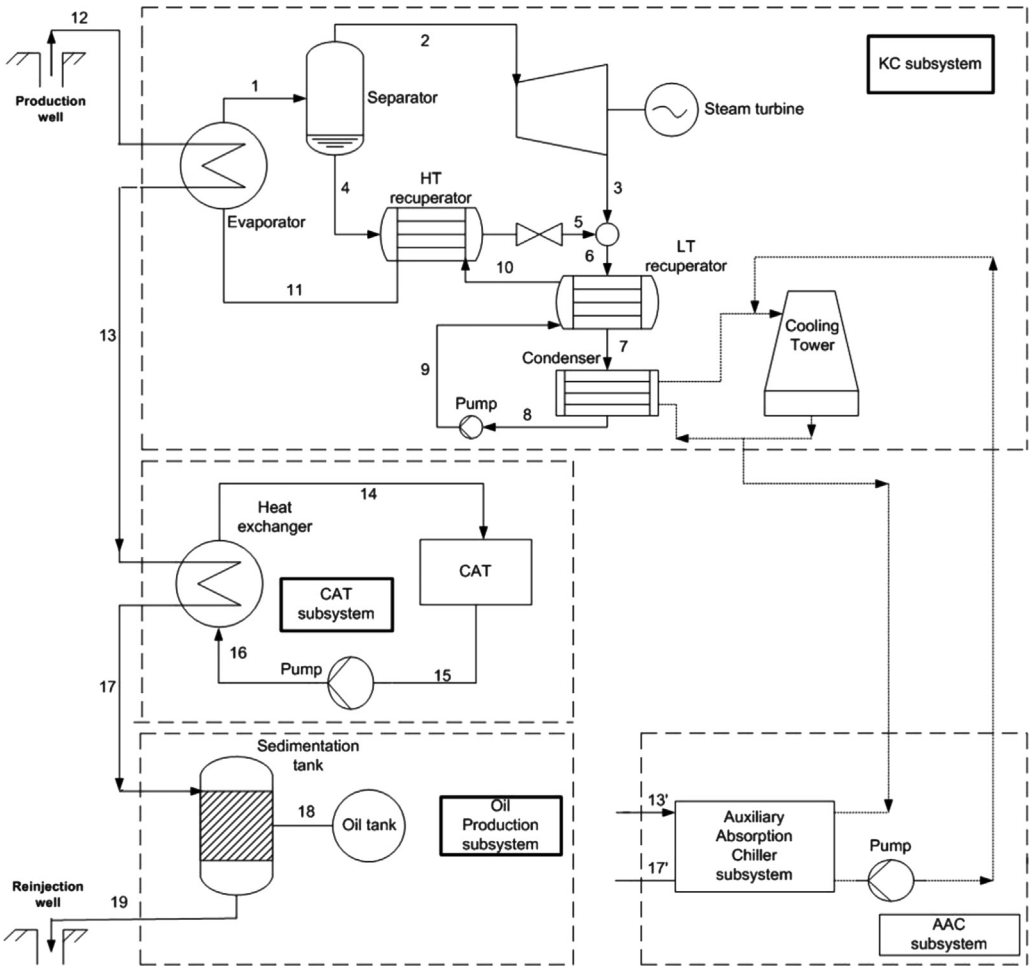


Fig. 11. System diagram based on KC cascade, from Ref. [39].

Another cascade utilization corresponds to the power generation coupling with processes in oil fields. In [37], there is a description of a power production system of low temperature by means of an ORC that interacts with the supply of heat directed to two oil processes. The feasibility of replacing the oil boilers and supplying the thermal energy by means of geothermal heat is analyzed. Fig. 8 shows the outline of the system. The geothermal resource, with a temperature of 110 °C, is used firstly to generate electricity and secondly to supply heat to the GHT (Gathering Heat Tracing) subsystem and then to the oil recovery subsystem. The study highlights that the proposed cascade utilization is economically profitable and is also technically feasible for its implementation in the processes of the oil fields.

Similarly, a performance evaluation of a Kalina cycle activated by geothermal energy and coupled with processes in oil fields [38] is presented. In this research, a mathematical model of the Kalina cycle is set and optimized by analyzing the thermal and exergy efficiency. Low temperature heat recovery is analyzed and the economic performance of the system of cascade utilization is evaluated. The results obtained show that the cascade utilization with electricity generation and recovery of crude oil is a way to expand the economic benefits of the facility. Fig. 9 shows the cascade system of two levels of temperature. In the first level, the Kalina cycle is activated by geothermal energy directly obtained from the geothermal well, and in the second temperature level, the subsystem of crude oil production is located. In order to increase the power of the KC, an absorption chiller has been incorporated to lower the temperature of the cooling water.

In order to determine the cascade system performance based on the KC and ORC cycles coupled with processes in oil fields, a comparative study was carried out in [39]. In the study, the ORC cycle performance and the Kalina cycle are numerically evaluated for different working fluids. It is demonstrated that the ORC subsystem performance using R236fa is better than other working fluids. Furthermore, the KC subsystem performance with 0.8 mass fraction of ammonia is the most appropriate alternative. However, for the conditions studied, the KC subsystem power is greater than that for ORC. In the research, geothermal resources of medium enthalpy are used with a temperature of 140 °C. In general, for both options, the use of geothermal water in oil fields can not only generate electricity and thermal energy, but also save large amounts of oil to increase annual profits. Fig. 10 shows the schematic diagram with the ORC, where the geothermal water is sent to the oil production subsystem after the electricity generation. In this case the subsystem CAT (Collection and Transport) for collection and transportation of oil, which consumes a large amount of thermal energy, is provided by a conventional boiler.

On the other hand, the schematic diagram of the cascade utilization based in the KC cycle is shown in Fig. 11. In this case, the geothermal fluid enters the KC evaporator to activate it, then passes to the CAT subsystem and later, to the sedimentation tank of oil production subsystem. Finally, the geothermal fluid is reinjected. The system also features the absorption chiller activated with geothermal energy.

Finally, regarding China, researchers from the Shanghai Jiaotong University (or Shanghai Jiao Tong University) show a small

experimental geothermal plant in cascade, with an ORC system coupled with an absorption chiller of two stages [40]. In the first level of the system it comprises an ORC that operates through geothermal energy and for the second level, it has the adsorption cycle. The plant diagram is shown in Fig. 12, and includes the two subsystems. The ORC operates with R245fa as working fluid, and the adsorption cycle operates with a mixture of CaCl2/BaCl2 in two stages.

Cycles are cascade connected through the heating fluid circuit flowing into the ORC first, and then, to the two stages of the adsorption chiller. For the process, geothermal heat is used twice sequentially, increasing the utilization efficiency of the geothermal heat. Thus, the generation of electricity and production of cold is achieved in a geothermal cascade process. The geothermal resource temperature is 98 °C (heat source), a temperature of 30 °C for cooling in the condenser and a temperature of −15 °C in the evaporator in the adsorption cycle. The experimental results show that it is possible to operate the plant with thermal energy in the temperature ranging from 78 °C to 98 °C; a proper temperature range for low enthalpy geothermal resources or waste heat from industrial processes.

4.3.2. Iran

In Iran, in the Sabalan geothermal region the artisan geothermal well of Gheynarjeh is located. It records a temperature of 83 °C and a flow of 7 l/s. In order to use this geothermal resource, two

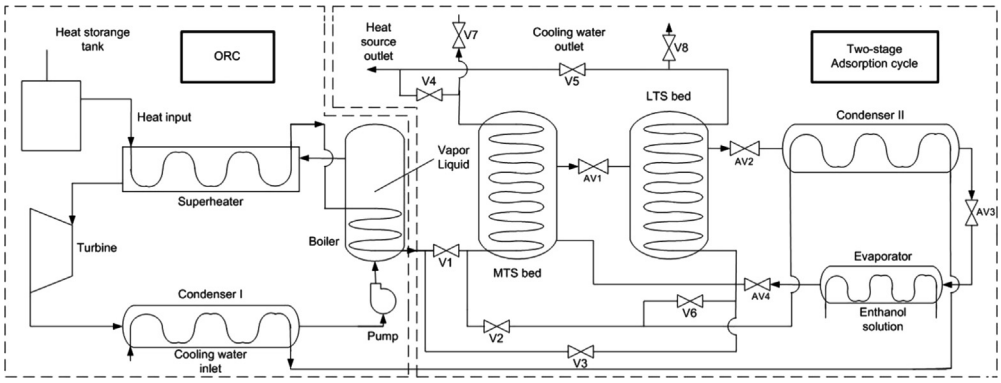


Fig. 12. Cascade production of electricity and cold, from Ref. [40].

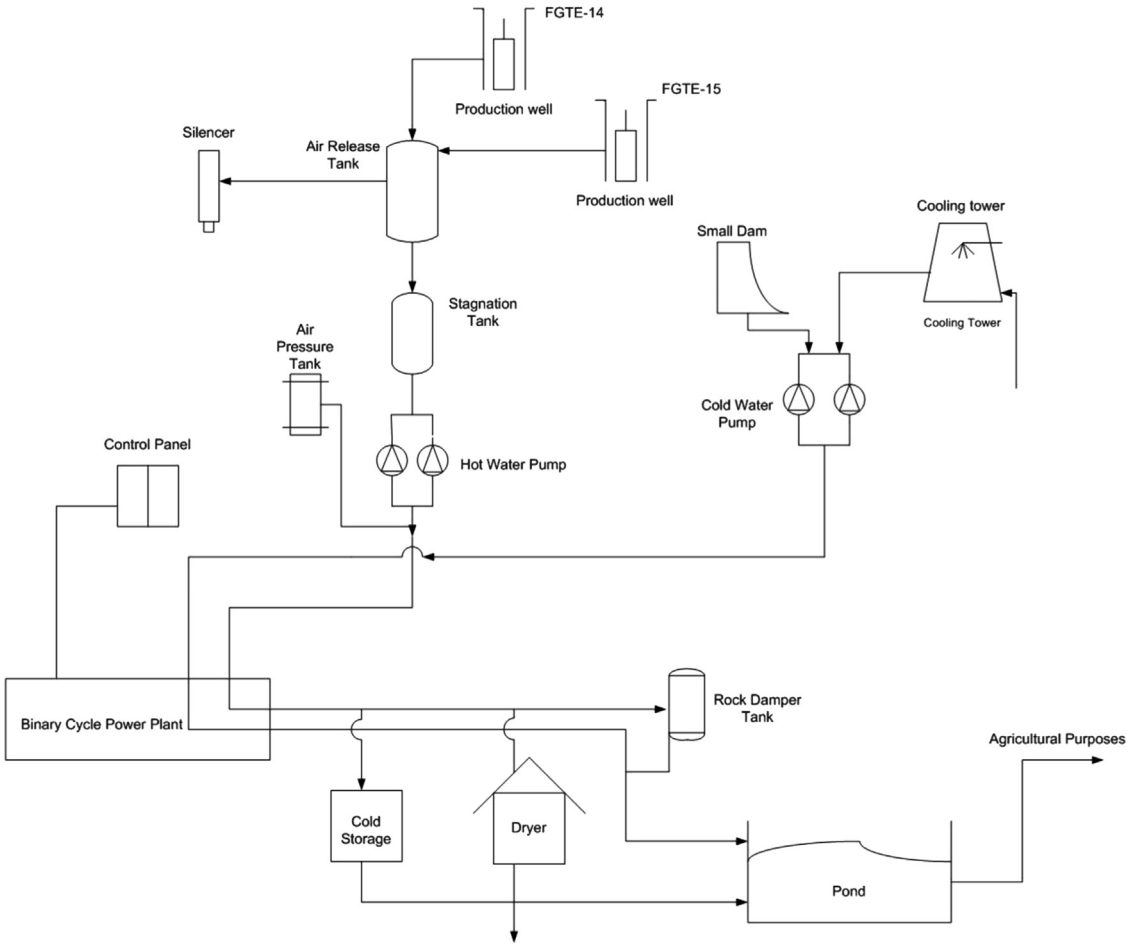


Fig. 13. Schematic diagram of the plant in cascade located in Thailand, adapted from [45].

alternatives are studied. The first alternative is to meet the needs of a pool, and the second raises the geothermal utilization for a district heating system. The analysis of the alternatives proposed is carried out with the condition of being later operated with a geothermal plant under the cascade utilization system. This work only mentions that the geothermal resource of the Gheynarjeh well is suitable to use in both alternatives and no more details are provided for the later use in cascade [41].

4.3.3. Thailand

In Thailand the EGAT (Electricity Generating Authority of Thailand) has installed a binary cycle plant of 300 kWe and has considered to use the residual heat for a dehydration process of chili pepper and garlic. The geothermal resource temperature entering the cycle is 116 °C and the geothermal water leaves the binary cycle at a temperature of 80 °C to circulate through heat exchangers installed in cabins of dehydration, [42,43]. The net power of the plant varies with the season from 150–250 kWe (175 kWe on average) producing 1.2 GWh per year. In this project other uses of geothermal energy are considered to dry agricultural products and for a Spa. In this case, the economic results show that this system is profitable. Fig. 13 shows the schematic diagram of the plant with the main subsystems and flows, [44,45].

4.3.4. Region summary

In the Asian region are encountered interesting proposals with cascade utilization of geothermal energy, all of them considering electricity production as the first level of the cascade system. The systems described for this region make use of medium enthalpy geothermal resource with temperatures as low as 83 °C suitable for electricity production in the first level of the cascade. The temperature of geothermal fluid for cascade systems in Asia range between 83 °C and 140 °C. According to the studies corresponding to China, it is possible to utilize oil wells as geothermal wells supplying heat for a cascade plant interacting with oil processes and replacing oil boilers. It is also advisable that cascade utilization is viable when the oil wells are in the high water-cut period.

On the other hand, environmental temperature is a key factor in the performance of cascade plants, such in the case of Thailand where the power output in summer season was decreased up to 50% increasing heat rejected for the second level. Therefore, if the application of second level cannot manage such amount of excess

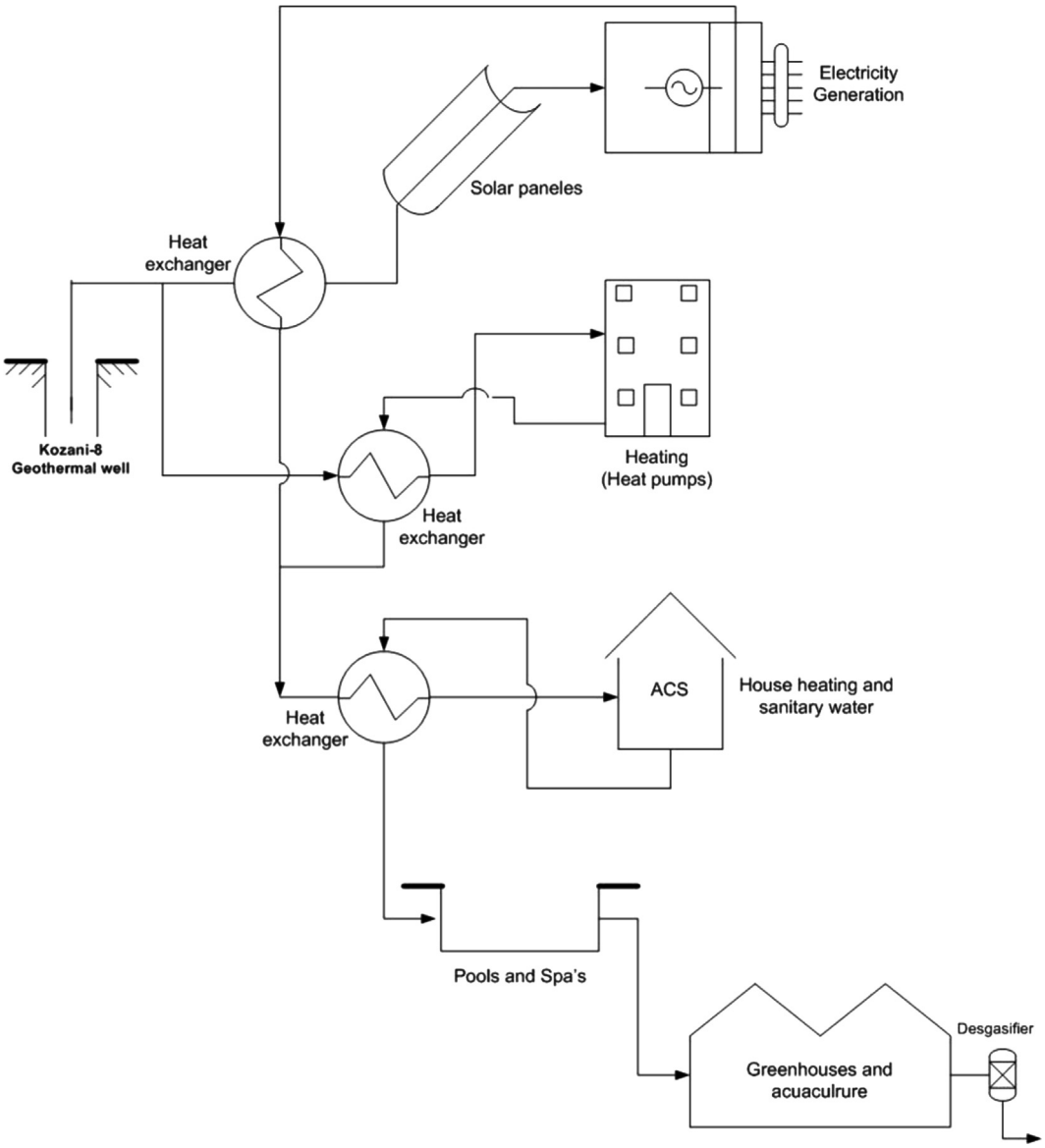


Fig. 14. Simplified proposal of cascade utilization in Albania, adapted from [54].

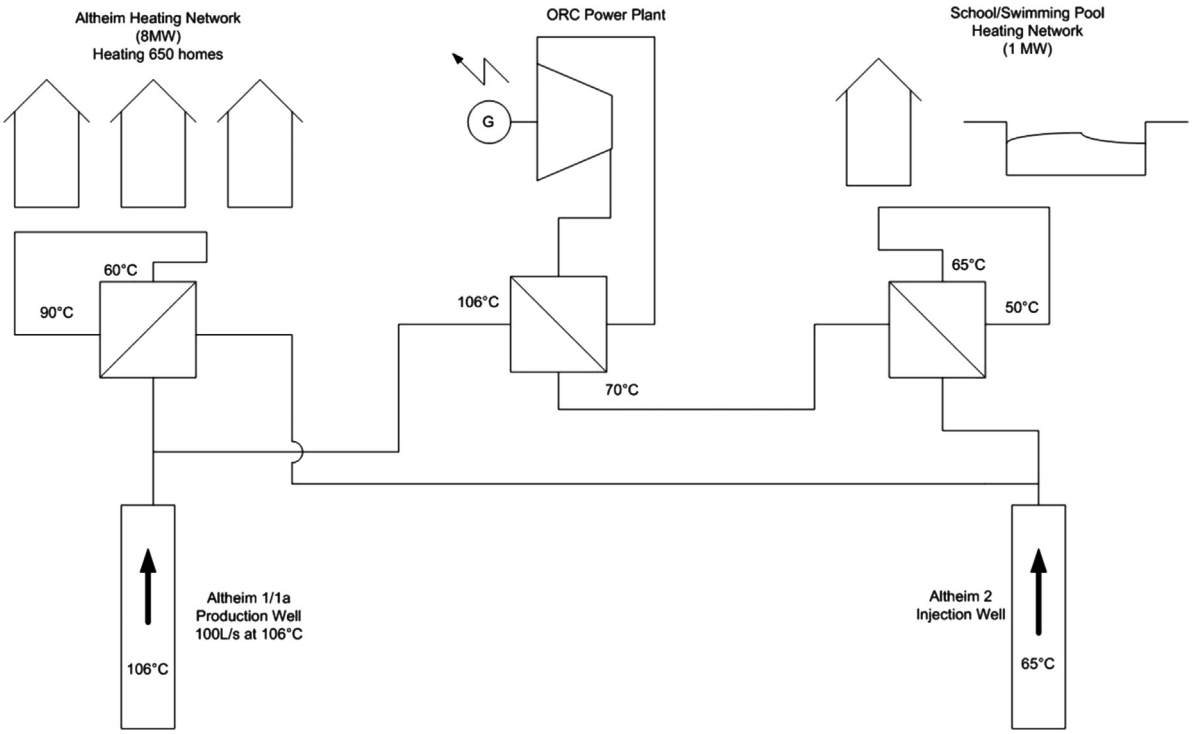


Fig. 15. Combined supply system of heat and power for the city of Altheim, sketched from [55].

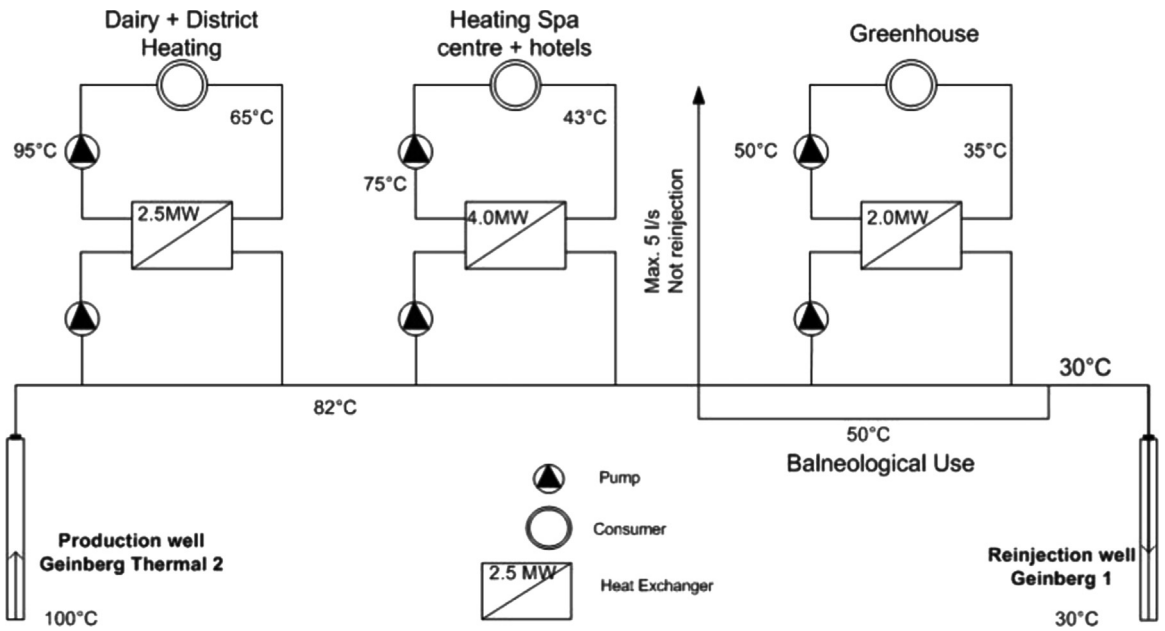


Fig. 16. Operating diagram of Geinberg, outlined from [56].

heat, the cascade overall performance is reduced severely. The main advantages of cascading geothermal energy in all the cases described for Asia are both economic and environmental.

4.4. Europe

Europe is undoubtedly the continent with the biggest geothermal applications in the world, at least in terms of direct utilization [46]. Among the countries that have been proposed multiple applications of geothermal energy under the cascade system are mainly: Albania, Austria, Greece, Iceland, Italy, Poland, Romania and Turkey. Romania widely reported several studies about cascade utilization to the University of Oradea. Iceland is the country with the greatest diversity of applications for geothermal energy, highlighting its power generation and its district heating system that supplies the town of Húsavik.

4.4.1. Albania

Albania is a country rich in geothermal resources of low enthalpy but despite this, it has not been possible to make massive use of these resources. This has motivated to research the integrated and cascade utilization of geothermal energy with low enthalpy as an effective system for the use of the geothermal resource and which represents an important economic profitability. The average temperature of the geothermal resource of the Albanian geothermal

springs is 60 °C, they are artisan springs with an average flow of 10 l/s. It has been observed that these resources are suitable for clinical Spa, supplying hot water and heating systems for hotels and resorts. Moreover, greenhouses for growing flowers and vegetables, and aquaculture facilities can be considered as well. This whole represents an opportunity to build resorts and service facilities operating with cascade utilization of geothermal energy [47–49].

Based on the above, there has been an innovative concept with a hybrid use of solar energy and geothermal energy in cascade, in which it is proposed what has been called a multicenter with various services such as spa, gym and thermal pools, including greenhouses and aquaculture. As a geothermal resource, the geothermal well Kozani-8 has been considered which records a temperature of 65.5 °C and a volumetric flow rate of 10 l/s, [50–54]. Fig. 14 shows the scheme of the multi-product system for the multicenter, it can be noted that the heating of the center is provided by the direct utilization of the geothermal energy (through the installation of heat exchangers), whereas for cooling, a geothermal heat pump is contemplated. The multicenter roof is covered with solar panels, which combined with geothermal energy can provide the thermal energy for an electricity generation plant and subsequently meet the needs of domestic hot water. This electricity will be used to supply the illumination system of the center under the concept of green energy. The study presents the economic analysis, based on the Net Present Value (NPV) with an estimated of 5.5 million Euros for total investment. Although investment is too high, the results of the economic study indicate that it is feasible. It is also mentioned that the project can significantly improve the quality of life of the community.

4.4.2. Austria

In the city of Altheim, a well was drilled to supply a district heating system in 1989. The well flow was 46 l/s at a temperature of 104 °C, equivalent to 10 MWth for district heating. The heating system has the capacity to supply heat to approximately 40% of the inhabitants of Altheim. 80% of the thermal energy is used to households, and the rest is used for the heating of a school and a swimming pool. A scheme of the power plant and the district heating is shown in Fig. 15 (Ref. [55]). In addition to the thermal energy, the geothermal fluid is also intended for the generation of electricity by an ORC with nominal power of 1 MWe. Once the geothermal fluid transfers its heat in the heat exchangers, it is reinjected at 65 °C. The district heating system operates about 1200 h per year at a cost around 0.05 USD/kWh (€ 0.04/kWh) based on a drop in water temperature of 30 °C [45,55].

The geothermal area of Geinberg has also been considered to implement a system that uses energy in cascade to obtain district heating, Spa in hotels, greenhouse heating and thermal energy for watering places. The system is activated by a geothermal resource at 100 °C and is composed of four thermal temperature levels. The geothermal fluid in the first level is used for supplying heat to dairy processes and district heating at a temperature of 95 °C, operating with a temperature difference of 30 °C. The second thermal level is used to heat a spa and a hotel heating at a temperature of 75 °C. The third thermal level is used for the greenhouse heating to a temperature of 50 °C. Finally, a fraction of the fluid is used for watering places and the rest of the geothermal fluid is reinjected. Fig. 16 shows the diagram for this system, [56].

4.4.3. Germany

Neustadt-Glewe is located in northern Germany, and there is a geothermal plant dating from 1995 to supply energy for district heating. The total output of the system is 11 MWth of which 6 MWth are produced by geothermal energy and the rest is provided by a natural gas boiler. In addition to producing useful

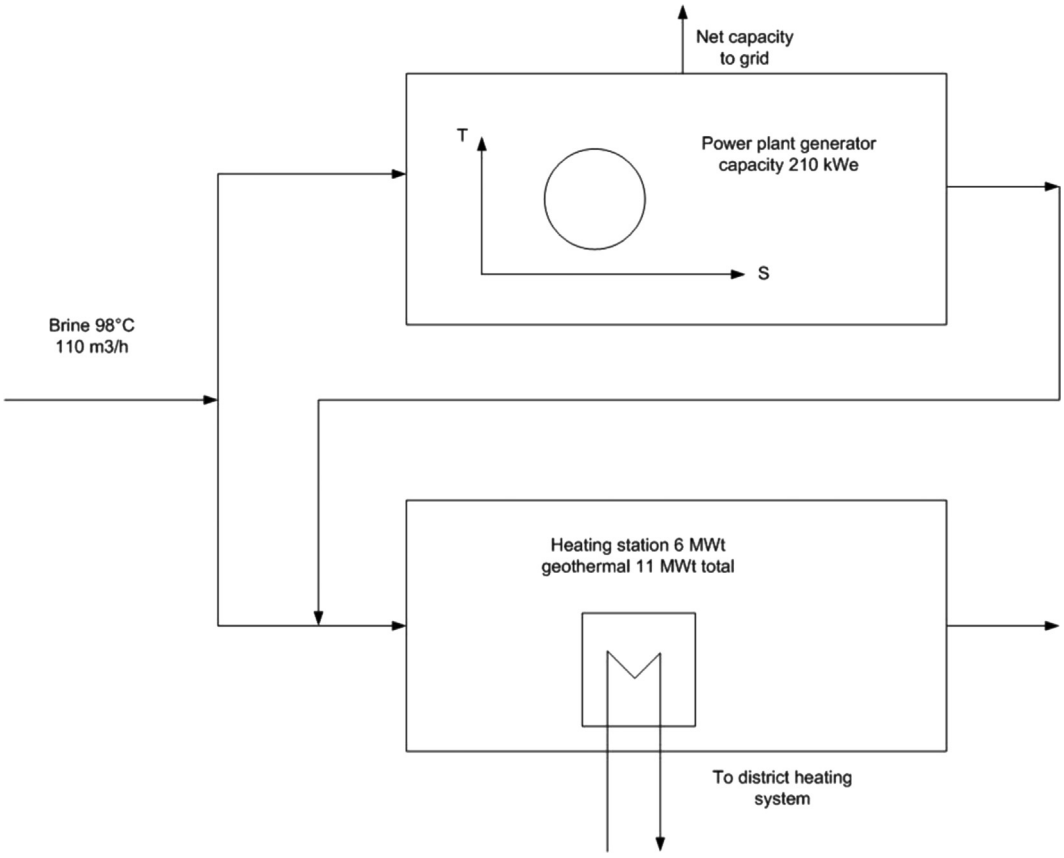


Fig. 17. Neustadt-Glewe geothermal plant to provide district heating and electricity generation, adapted from [57].

thermal energy, the plant also produces electricity through a 210 kWe ORC with n-Perfluoropentane (C_5F_{12}) as the working fluid. For the generation of both products there is a geothermal fluid with a temperature of 98 °C at a rate of 30.6 l/s (See Fig. 17, Ref. [57]). Due to varying temperature conditions in summer and winter, the power plant operates under varying flow conditions at constant temperature. After being used, the geothermal fluid is re-injected at a temperature of around 50 °C. The total cost of the district heating project was 12.3 million USD with 1.24 million USD for the power plant [45,57].

4.4.4. Greece

In 2002, a project was proposed by the Community Research and Development Information Service (CORDIS) for cascade utilization of the energy in the geothermal field of Xanthi. They propose the use of geothermal energy for pasteurization of milk, in parallel with the production of cold through absorption chillers for district cooling. After the pasteurization process, the geothermal energy will be used for greenhouses heating and cold storage. Geothermal wells are approximately 2500 miles away from the applications. The geothermal fluid transfers its heat to the different uses, and then, it is reinjected. The district network intents to supply the energy requirements of heating and cooling for schools, churches, the town hall, the center of information for farmers, etc. For the pasteurization process, an inlet temperature of 75 °C and an outlet temperature of 45 °C is required. For greenhouses crops, it is proposed to produce 60 t per year, maintaining an internal temperature of 14 °C. The proposed system highlights cascade utilization as a means of protecting the environment by significantly reducing the emission of pollutants and supplying sustainable energy [58].

Another case reported for Greece, corresponds to the district heating project that consist of using heat pumps operating with geothermal water with temperatures between 20 °C and 40 °C. The project is located in the city of Langadas near Thessaloniki. It includes the heating of a hotel with Spa, an elementary school, a secondary and a health center heating, and a new city hall. The scheme can be seen in Fig. 18 (Ref. [59]). The project is in development and it was supported by the THERMIE program [59].

4.4.5. Hungary

In Hungary, under the Geothermal Communities program of the European Community, it is investigated the technical and economic feasibility to replace natural gas boilers in the district heating network that supplies various public buildings for the small town of Morahalom. The replacement considers the direct utilization of geothermal heat in cascade to different temperature levels and the incorporation of a heat pump. In the project, it is considered using a geothermal well with a depth of 1260 m, a flow of 25 l/s and a wellhead temperature of 62.1 °C. To implement the project a total of 2.05 million Euros was allocated. More information regarding this case is not reported [60,61].

4.4.6. Iceland

Iceland is a country that has significantly used geothermal resources, mainly the direct utilization for heating. Particularly in Husavik, they began to use geothermal energy for heating. Later, because of the high geothermal potential available, it was decided to expand the scope of geothermal use for electricity generation [62]. Fig. 19 shows, on the left side, the system only with the direct utilization for heating, and on the right side the system with electricity generation.

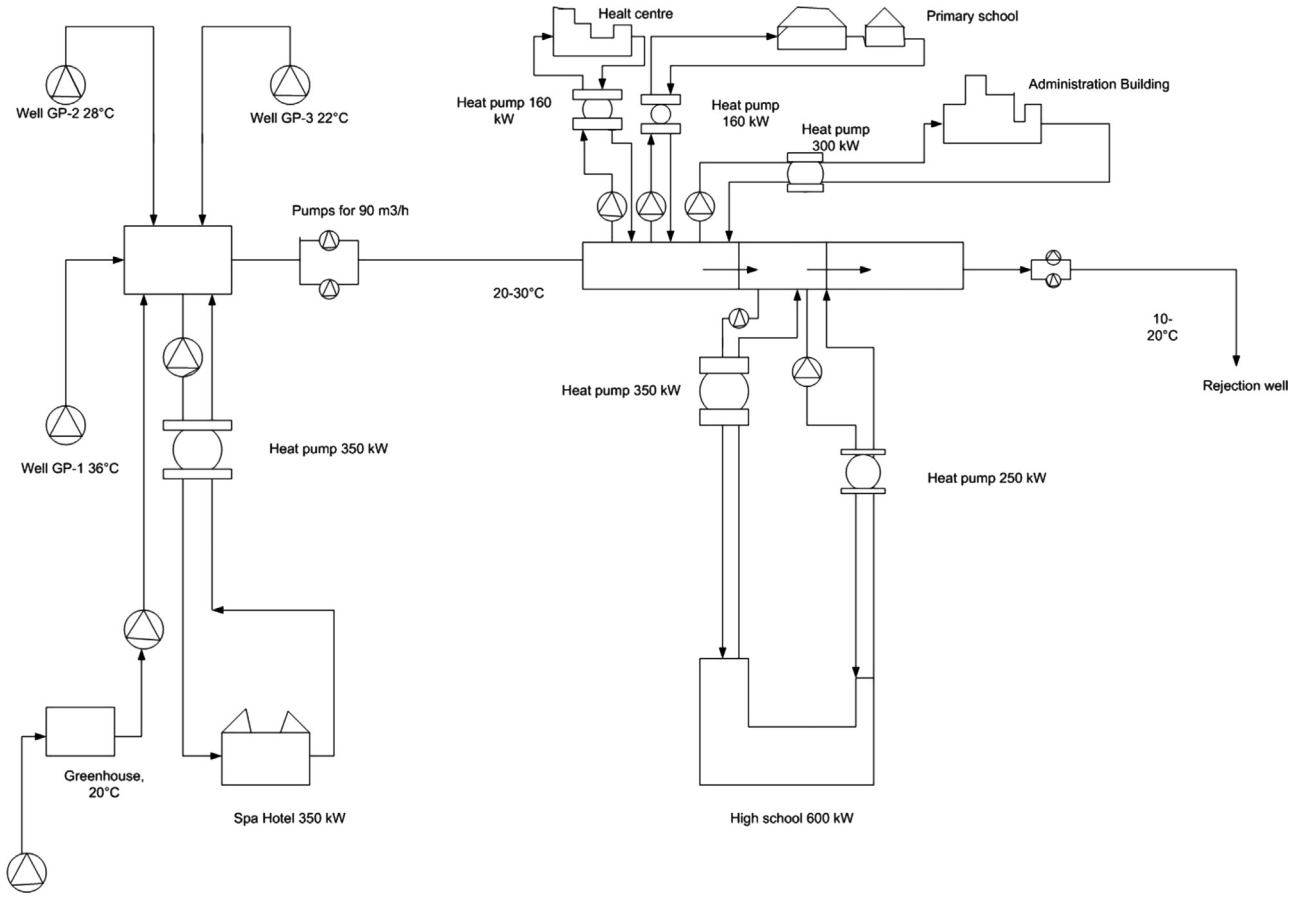


Fig. 18. Scheme of cascade utilization using heat pumps, sketched from [59].

It is formally proposed, from the extended system of heating and power generation, an integrated plant scheme for multiple uses of geothermal energy by using geothermal resources with medium and low enthalpy [62–64]. For the production of electricity, using binary cycle and ORC was considered in addition to what at the time was cutting edge technology, the Kalina cycle. Finally, it was decided to install the Kalina cycle with a nominal power of 2 MWe to operate with geothermal resource of 124 °C. However, the plant generates 1.7 MWe because the actual temperature is 121 °C. In this scheme of multipurpose utilization, the

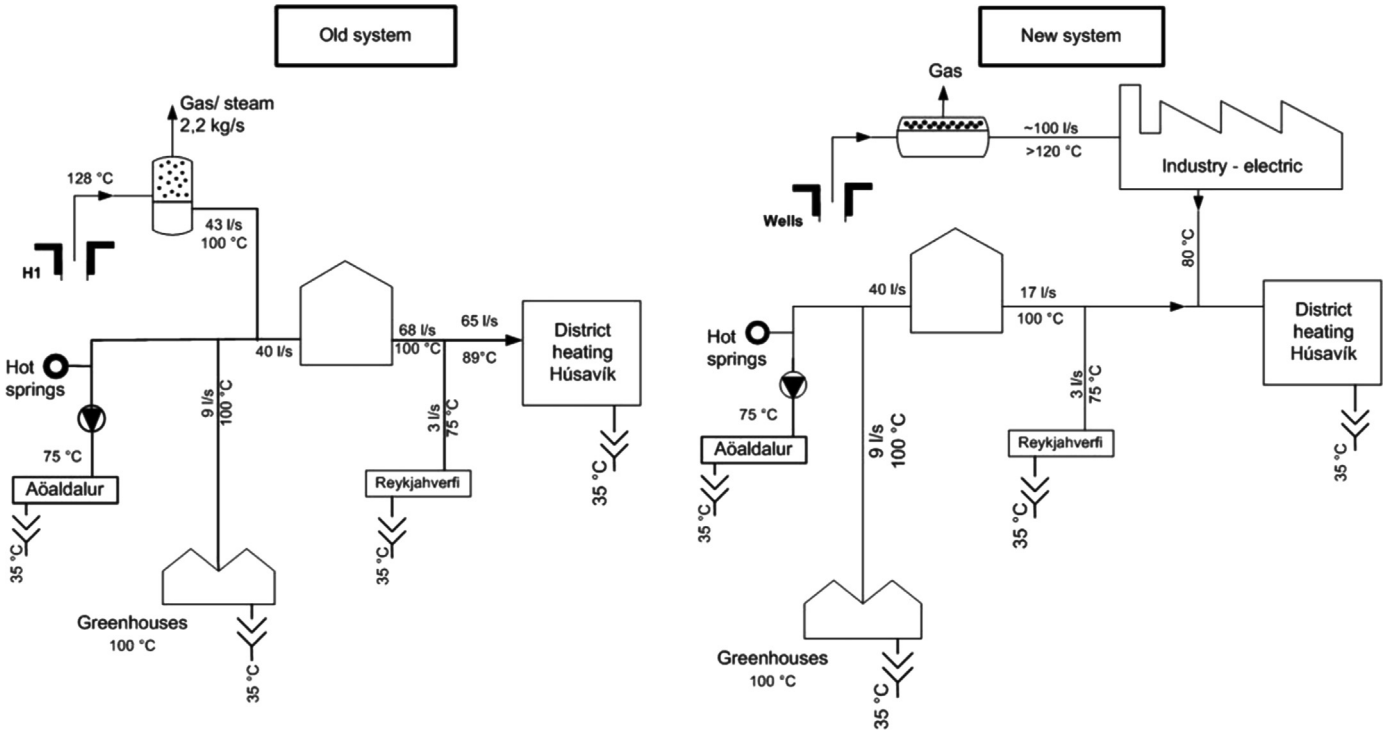


Fig. 19. Extended use of geothermal heat for Husavik, sketched from [62].

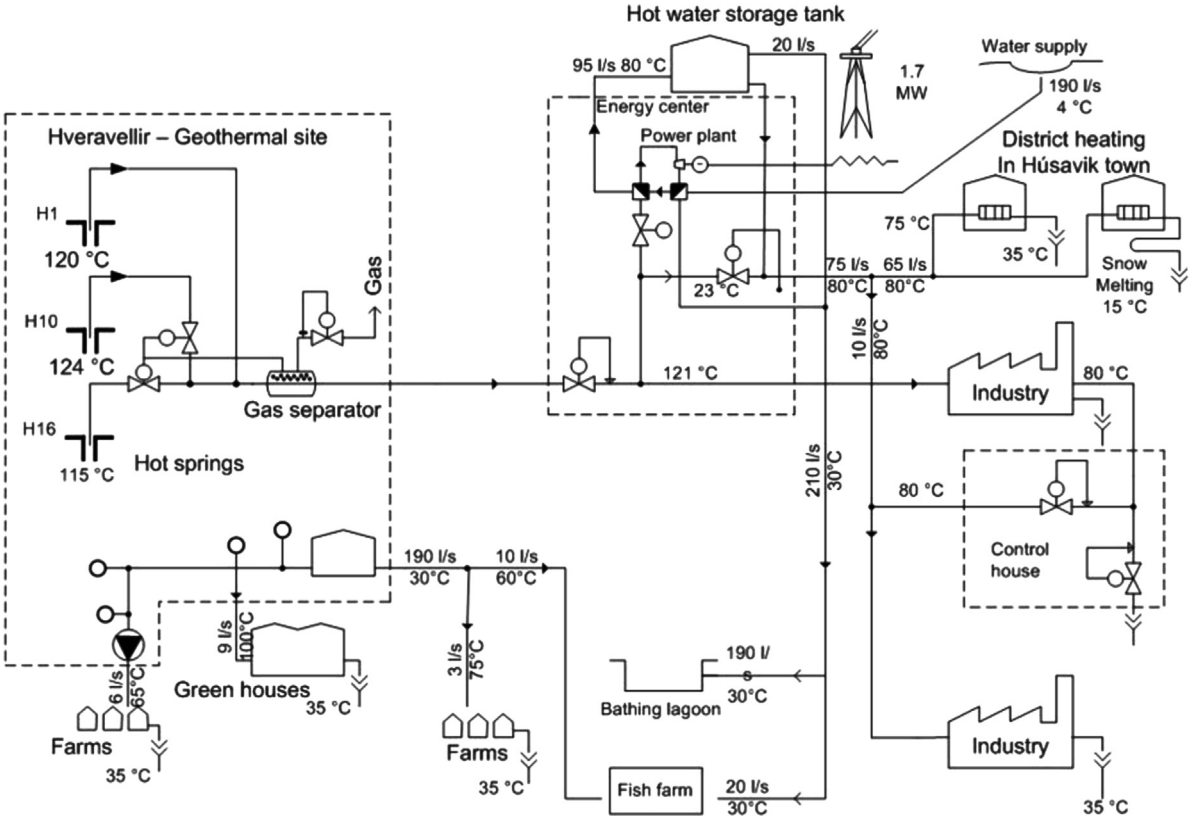


Fig. 20. Húsavik diagram of the multiple uses of the geothermal energy, adapted from [63].

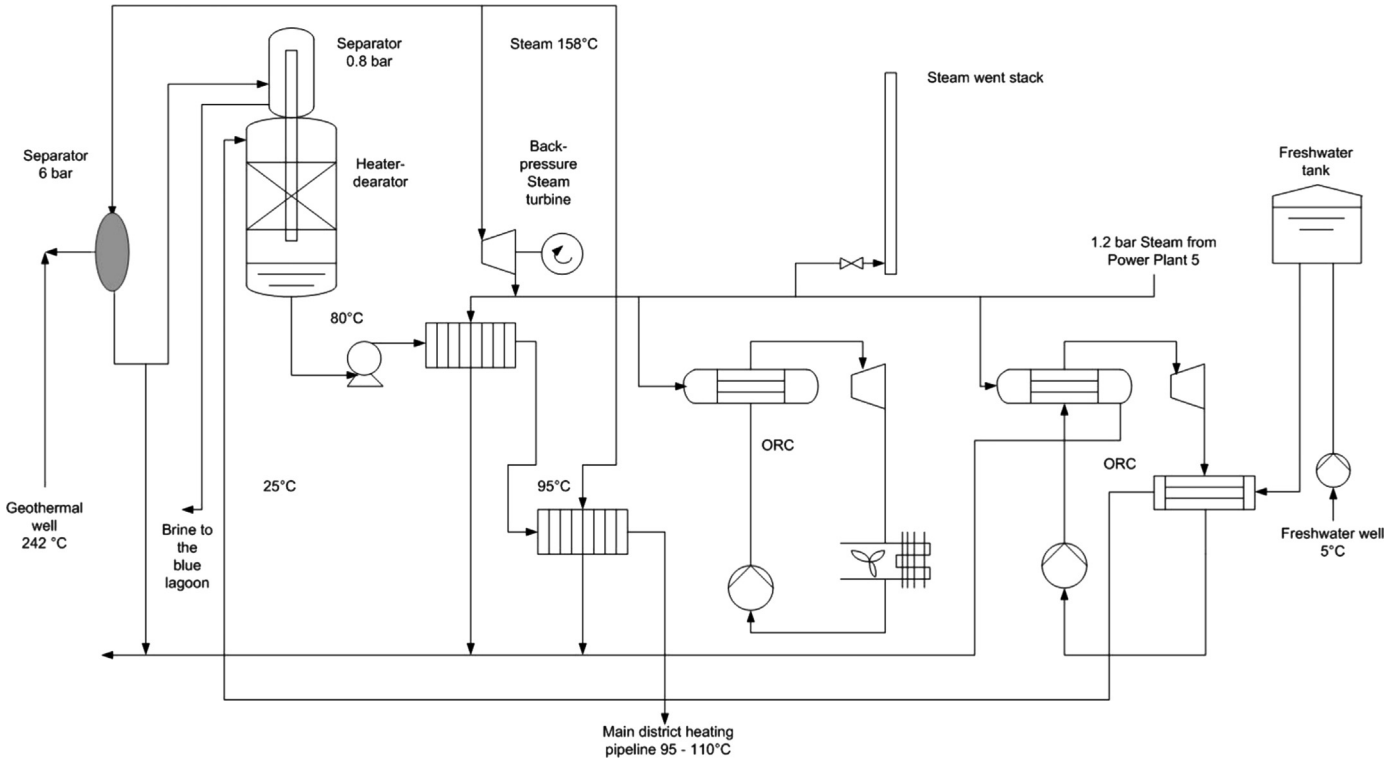


Fig. 21. Flowchart of the Svartsengi Power Plant, adapted from Ref. [65].

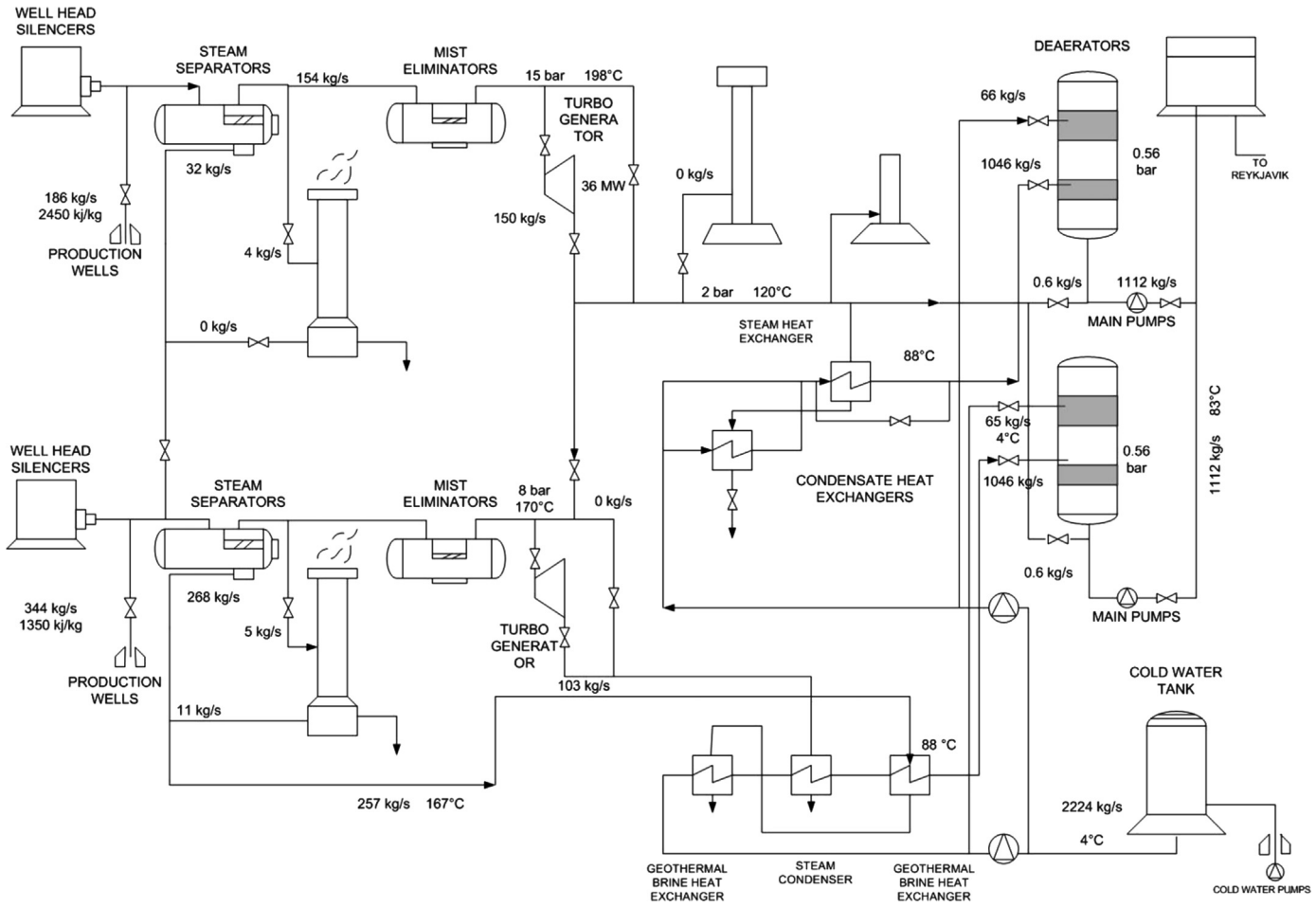


Fig. 22. Nesjavellir plant which supplies electricity and district heating to the city of Reykjavik, sketched from [66].

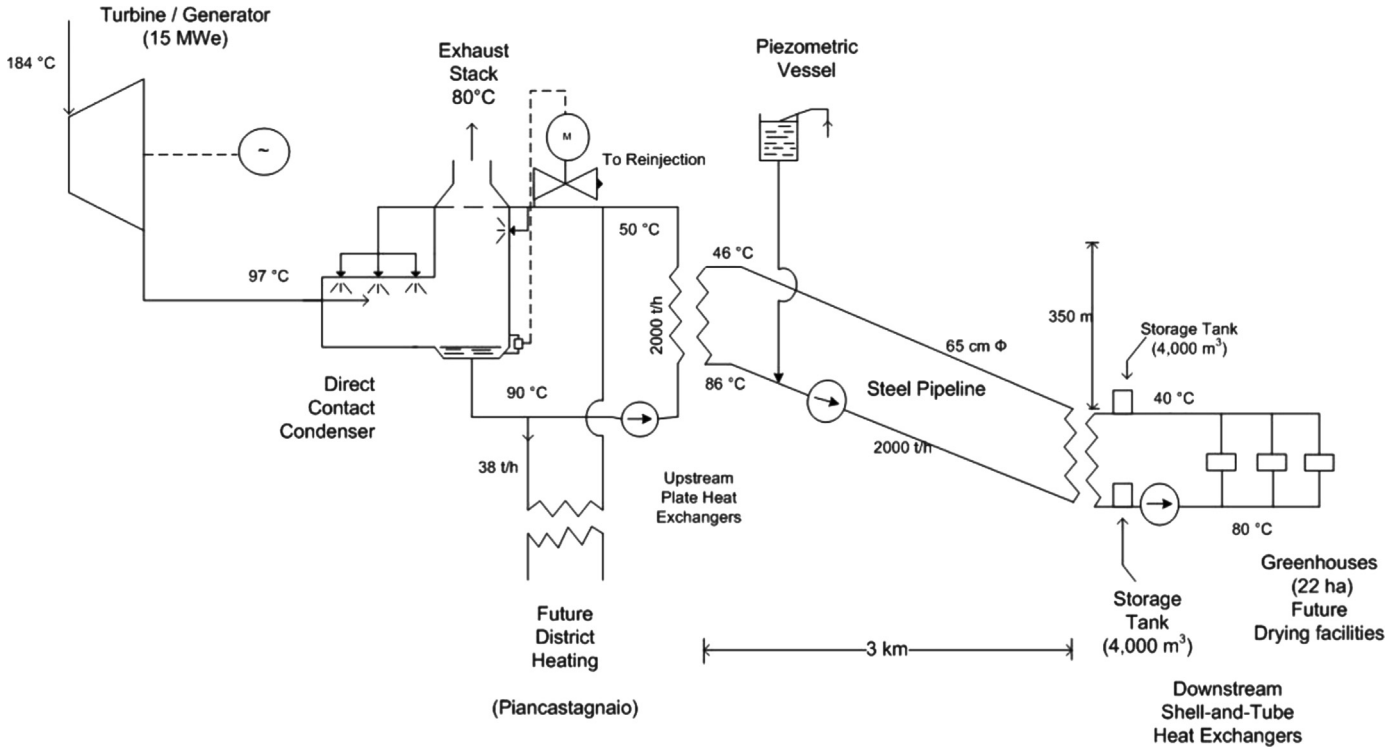


Fig. 23. Scheme of heat recovery at the power plant, sketched from [67].

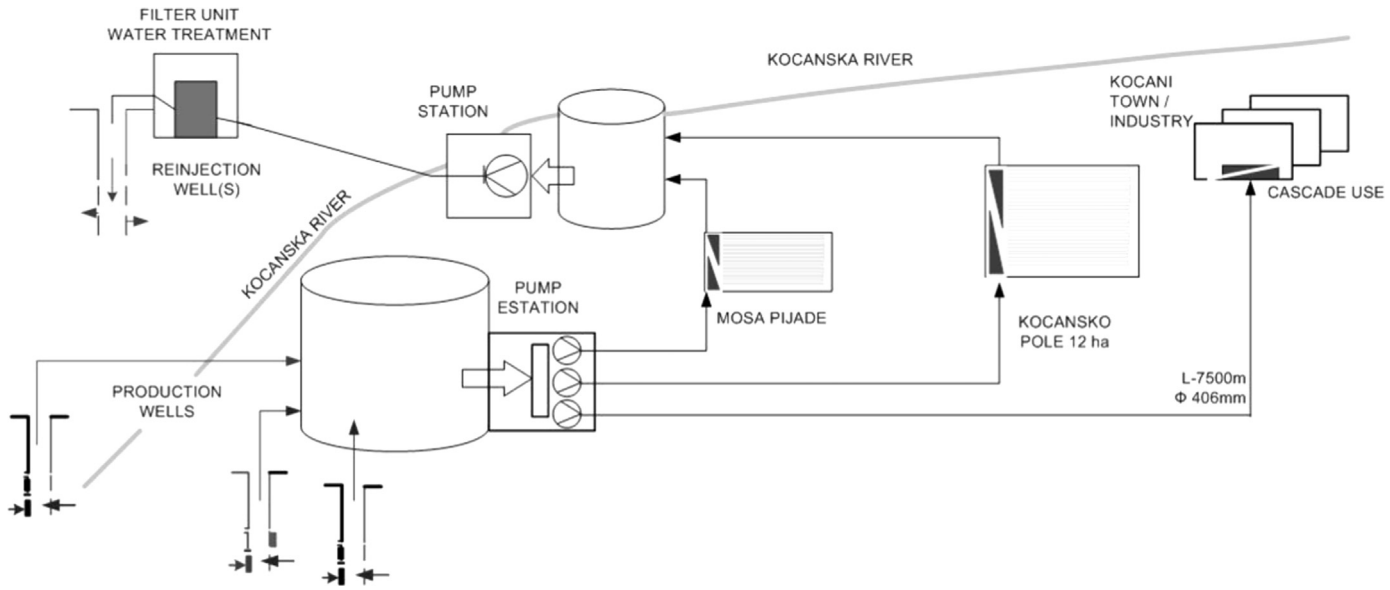


Fig. 24. Scheme with the incorporation of cascade utilization of geothermal water, outlined from [68].

heat supply is considered for industrial processes, farms and greenhouses, a fish farm, melting ice, and a recreational use in a thermal lagoon. Fig. 20 shows the uses and temperature levels. The total project cost is about 12 million Euros, from which 8 million were spent for retrofitting of the district heating network and 4 million to the power plant. Among the improvements in progress is wood drying, production of glucosamine and tilapia hatchery.

In Iceland, another application with a simultaneous production of electricity and thermal energy is found in the Svartsengi plant, which supplies hot water to a district heating system to Reykjanes. It provides electricity to Hafnarfjordur and other communities. The total generation capacity of the plant is 46.4 MWe and 200 MWth (2700 TJ/year) in the form of hot water for the district heating. The plant is located near the city of Grindavik to the southwest of Reykjavik. The system has 12 production wells and one for reinjection. The reservoir fluid is a brine at 240 °C. The remaining hot water flows into the adjacent lava field and the famous bathing area “Blue Lagoon” [45,65]. A flowchart of the Svartsengi Power Plant is shown in Fig. 21.

Another similar application corresponds to the Nesjavellir plant found on the west of the Lake Thingvalla, which supplies electricity and district heating to the city of Reykjavik. This field reaches temperatures of around 380 °C. The main purpose of the

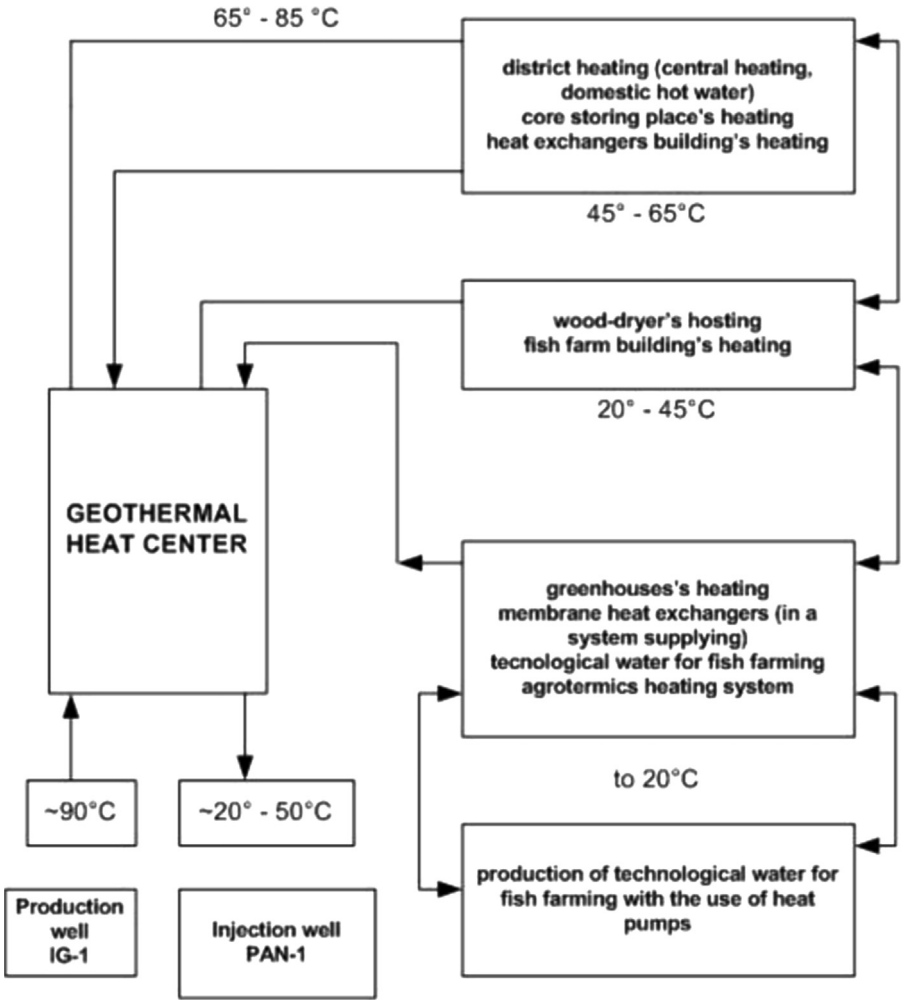


Fig. 25. Utilization system of geothermal energy in the Geothermal Laboratory PAS MEERI, sketched from Ref. [69].

plant is to provide hot water to the Reykjavik area. The capacity of the plant is of 270 MWe and 290 MWth equivalent to 4500 TJ/year at a rate of 1100 l/s to 83 °C for the city of Reykjavik. The plant has a total of 22 drilled wells to depths ranging from 1000 to 2200 m. A flowchart of the plant is shown in Fig. 22. In this installation the pipeline that leads the hot water to the city of Reykjavik has a length of 27 km, with dilatation points every 200 m, and it is able to carry water at 96 °C at a rate of 1.870 l/s. Water takes about seven hours to travel the distance, losing only 2 °C, [45,66].

4.4.7. *Italy*

The literature reports, for Italy, an installation that uses geothermal energy for the Monte Amiata geothermal area. In the first place, this installation produces electricity and then supplies heat for district heating, greenhouses and a dehydration process. The plant produces 15 MWe of electrical power from geothermal steam at 184 °C, 9.4 bar and a mass flow of 55 l/s (198 t/h). The exhaust steam passes into a condenser of direct contact where the energy unutilized is recovered by means of heat exchangers, which provides thermal energy at 80 °C for greenhouses and the dehydration process. For the district heating system, a temperature supply of 90 °C is considered [67]. Fig. 23, sketched from Ref. [67], shows the flowchart for the facility of Monte Amiata project. This project highlights that the facility is a good example of cascade utilization of geothermal energy that can help the local economy.

4.4.8. *Macedonia*

For the people of Kocani, located in the northeastern part of Macedonia, the use of geothermal hot water has been proposed, mainly for district heating and greenhouses, [68]. For this location, the use of geothermal resources by improving the existing geothermal system was maximized. This system reported environmental problems due to the discharge of thermal water into the Kokanska River, which caused the increase in temperature of the river water, and thus affecting the natural water cycle. Among the alternative solutions, cascade utilization of the geothermal resource is proposed to reduce the discharge temperature with better energy efficiency. Fig. 24 shows an outline for the proposal. For the system, there is an outlet temperature of the geothermal wells between 73 °C and 78 °C and a maximum flow of 450 l/s. Among the benefits of the cascade, it is highlighted the system efficiency, the diminishing of the environmental impact and an improvement of the community economy [68].

4.4.9. *Poland*

In the Geothermal Laboratory Pas Meeri located in Podhale, a pilot project is documented in which cascade utilization of geothermal energy is implemented. The project involves the implementation of a heat recovery system in cascade for fish hatchery and vegetables growing. Fig. 25 shows the scheme of the pilot project [69]. The system considers an outlet temperature of the geothermal well between 80 °C and 90 °C to enter the geothermal power central

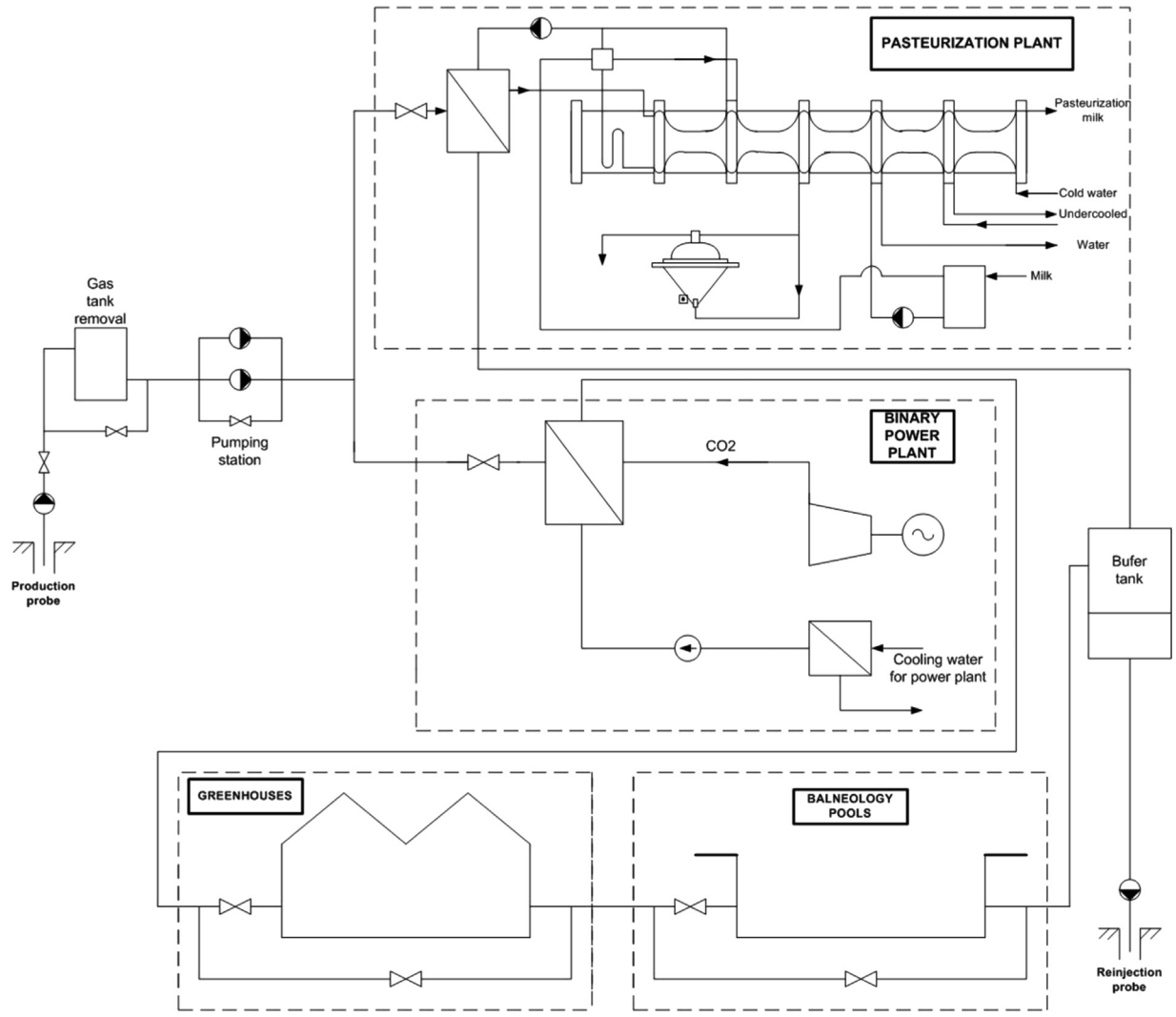


Fig. 26. Diagram of multiple use of geothermal energy in Romania, sketched from [71].

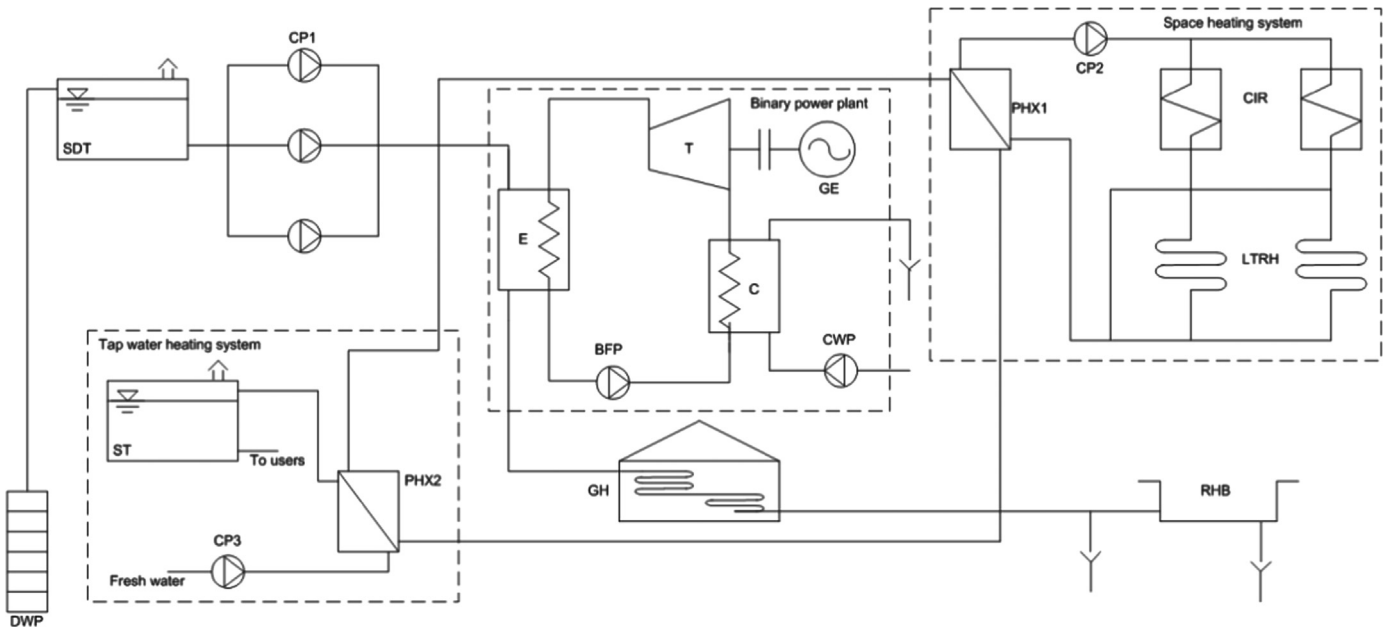


Fig. 27. Cascade utilization of geothermal water layout at the University of Oradea, sketched from [72].

and to provide hot water for district heating for apartments and buildings and hot water for domestic use. After the first thermal utilization, water has a temperature between 45 °C and 65 °C, which is enough temperature to breed fish and to grow vegetables. Finally, water is reinjected. The proposed system provides data which validate the increased efficiency and profitability of the plant.

A patent is also presented for Poland, wherein a method for using heat sources of low and medium temperature is described [70]. The claimed invention is characterized by its use for pre-heating the working fluid in a plant comprising two thermally coupled ORC through a heat exchanger. Its description is virtually summarized as ORC cycles that can be interconnected in cascade

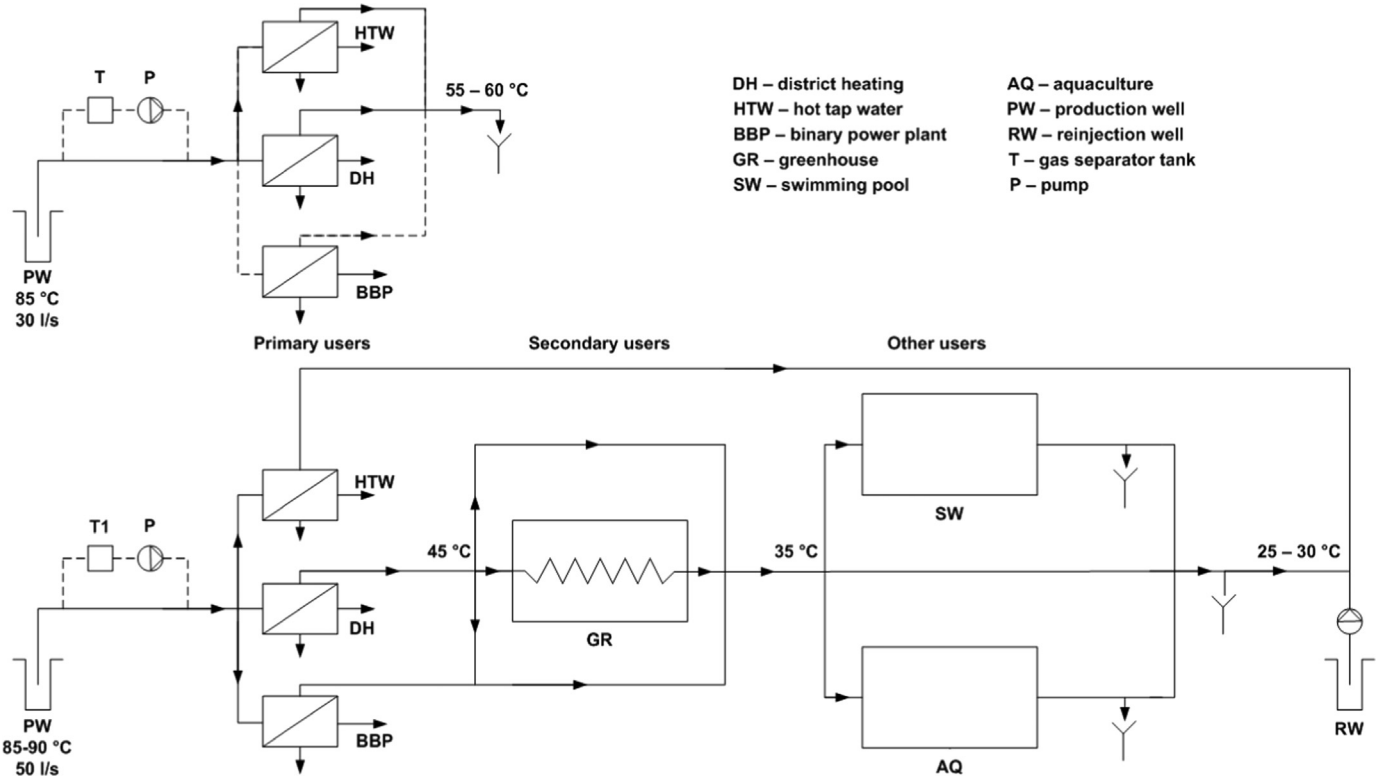


Fig. 28. Diagram of the existing facility and perspectives, outlined from [73].

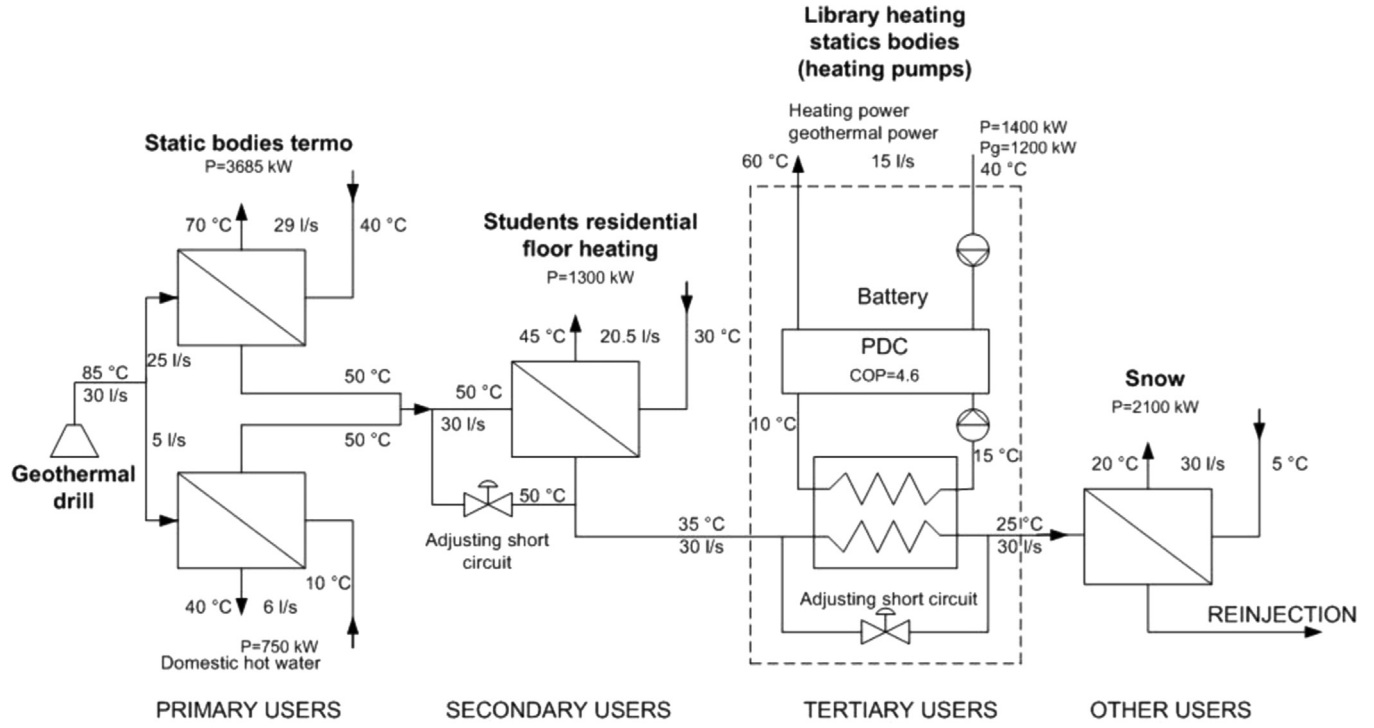


Fig. 29. Block diagram of the thermal center of the pilot plant using heat pumps, sketched from [74].

via heat exchangers. Possible interconnections can be performed on turbine extractions and, through the heat rejected in the condenser of the higher temperature ORC. In this system only electricity is generated at two temperature levels.

4.4.10. Romania

Researchers at the University of Oradea, Romania proposed to use geothermal energy for electricity production, pasteurization of milk, supplying heat to a housing and balneology [71]. For the system proposal is considered a temperature of 100 °C of geothermal water, with a flow of 60 l/s and a pressure of 4 bar. The hot water from the well is separated into two streams, one for the pasteurization of milk and one for generating electricity through a binary cycle. Geothermal water leaving the evaporator of the binary cycle is directed to greenhouses and subsequently balneology pools. The stream at the output of the pools joins the discharge of the pasteurization process, and it is finally reinjected. This cascade utilization is shown in Fig. 26.

Another study that considers the possibility of covering the thermal needs of the campus of Oradea by hot geothermal water is presented in [72]. Within this campus is a well of 2991 m, which was drilled in 1981 and currently reports a hot water flow of 25 l/s and a temperature of 84 °C. Initially, the well was designed to supply hot water and heating to the campus and three apartment blocks. To maximize the use of the geothermal resource, it was proposed to incorporate the system with a binary cycle plant with CO_2 as working fluid, a greenhouse and a pool. Figs. 27 and 28 show the proposed arrangement with cascade utilization of the energy for the campus of the University of Oradea. In this cascade system geothermal water from the well is separated to feed the binary cycle and meet the domestic hot water and heating needs at the same temperature level. Afterwards, the water leaving the evaporator of the binary cycle, which has a temperature between 45 °C and 50 °C, is directed to the greenhouse and the last level of temperature to the pools. Finally, the geothermal water is reinjected. It is important to point out that due to the complexity of the system, programmable logic controllers (PLC) and a system control and data acquisition (SCADA) have been used as tools to determine the best operating strategy. The technical calculations of the study demonstrate that the proposal presents energy savings and a reduction of CO_2 emissions. A full report with all the information regarding this project can be found in [73].

In regard to the cascade proposal for the campus of Oradea, another study considers the use of heat pumps to supply underfloor radiant heating to other buildings, as well as a domestic hot water supply. As the last level of the utilization of the geothermal water, thermal energy is proposed to be used for melting snow [74]. Fig. 29 shows the scheme for cascade utilization in this proposal.

Another cascade utilization has been recently documented for the district of Iosia in Oradea, [75]. For this system, the energy resource comes from three geothermal wells which total flow is 29 l/s with an average temperature of 105 °C. The flow of one of the wells will be used for feeding an ORC able to produce 50 kWe of power, the working fluid of the unit is HFC245fa, which does not

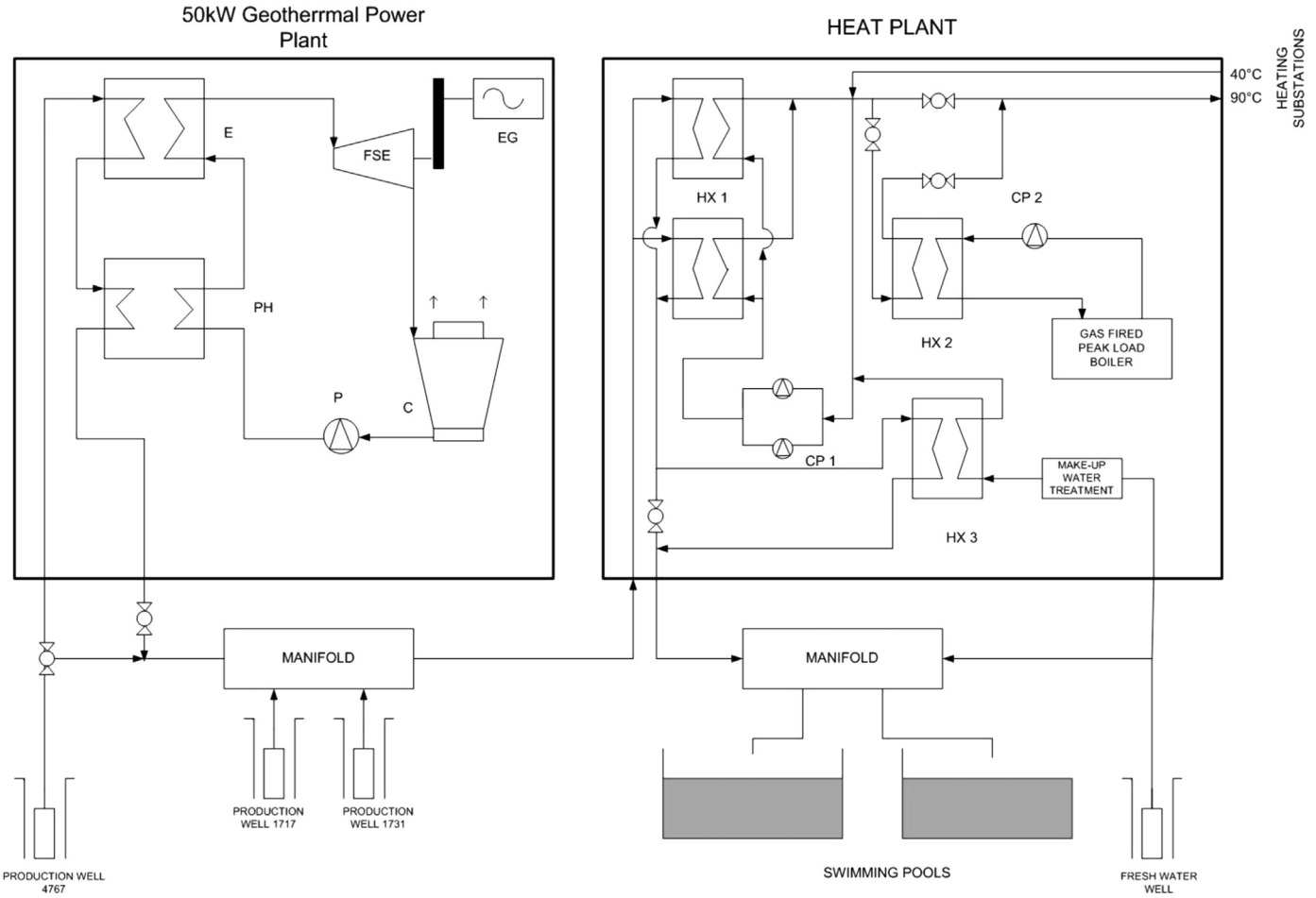


Fig. 30. Cascade scheme for the district of Iosia, Oradea, sketched from [75].

damage the ozone layer. The flow at the exit of the ORC joins the flow of the other two wells in a wellhead, and subsequently, it is sent to the plant of thermal energy services to provide heat to the district heating network and other public buildings. In the last level of the cascade, there are a few swimming pools. The system has auxiliary boilers mainly to cover the thermal demand peaks. Fig. 30 shows a scheme for this facility. Operation data for this plant show that cascade scheme has increased the factor of geothermal resource utilization [75].

4.4.11. Slovenia

For this country, cascade utilization is reported for the city of Lendava. It is mentioned that geothermal energy is used in cascade manner to produce simultaneously electricity, district heating and cooling. Other direct uses are also considered such as, aquaculture, greenhouse heating and Spas. The temperature of the geothermal resource for this region is reported at 80 °C [76].

4.4.12. Turkey

In Turkey, a research has been conducted on the integrated utilization of geothermal resources, considering the conditions of the geothermal region of Kutahya, Simav. This region reports ten geothermal wells with depths between 169 and 765 m and temperatures from 84 °C to 162 °C, these are suitable temperatures for power generation, heating and hydrotherapy. In this regard, a multi-purpose complex has been proposed to use geothermal energy more efficiently. For the generation of electricity, the feasibility of integrating a binary cycle that later uses a geothermal fluid has been investigated to provide energy as heat for homes and greenhouses, and a Spa. Fig. 31, sketched from Ref. [77], shows the schematic diagram of the system. The exergy analysis and life cycle cost have been used to carry out the research. The results show that the multiple use of geothermal energy is an economically profitable option for the conditions of the geothermal region of Simav [77].

4.4.13. Region summary

Due to its geographic and climate characteristics Europe has undoubtedly the most wide variety of applications for cascade use of medium and low enthalpy geothermal energy, highlighting power generation, direct and recreational uses, industrial processes, district heating, space heating, hot water and heat pumps.

The power generation is the most used application, used in 63% of the total of this type of systems. Main direct uses implemented in European cascade systems are heating networks, greenhouses and balneology, due to the particular needs of the continent. Currently, geothermal district heating has a share of about 58% of

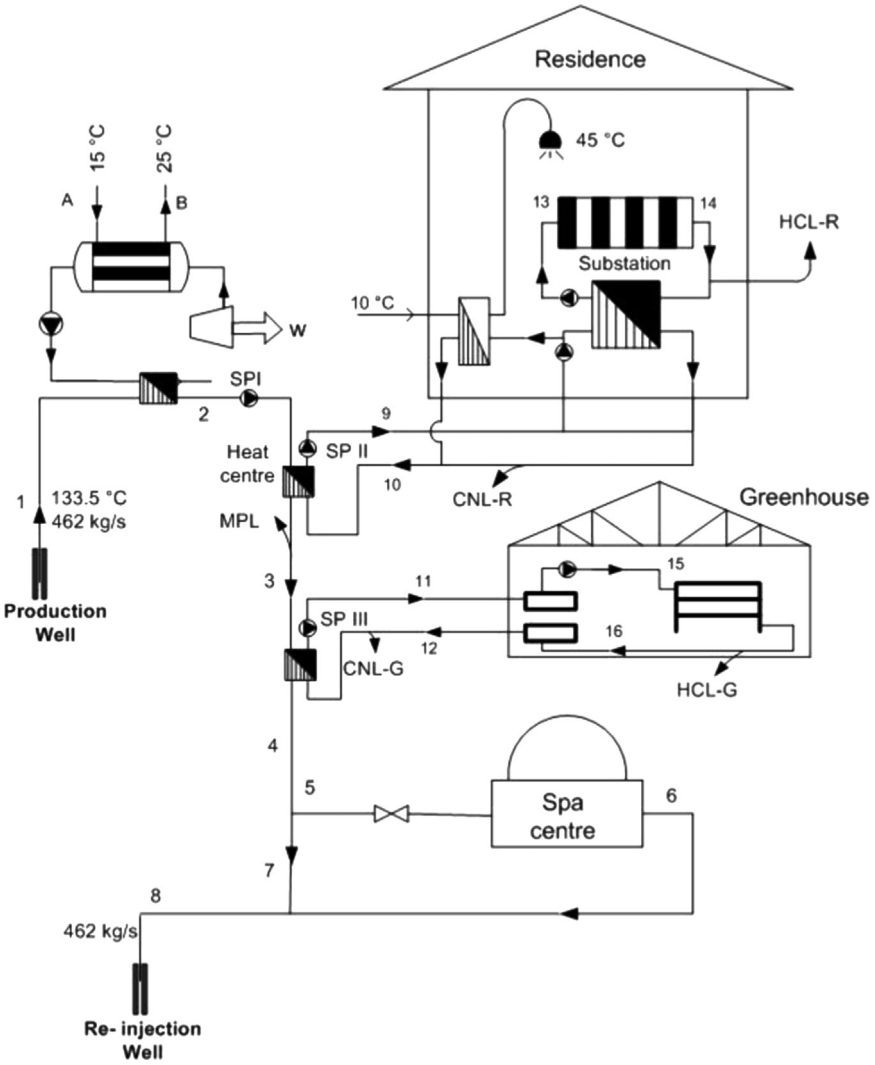


Fig. 31. Integrated system by geothermal resource, sketched from [77].

the total European cascade solutions. The countries that show greatest cascade use are Iceland, Austria, Greece and Romania.

An advantage of the cascade systems for this continent is that have a wider temperature range of operation, because the average environmental temperature, much lower compared to other regions. It is worth mentioning that the implementation of these technologies provides better quality use of energy in its counterpart with higher investment costs. Other feature for this continent is the advanced technological development and continuous research allowing the implementation of cascade systems for residential sector, industry, government and public services and agriculture. The living standards also allow focusing more on recreation and leisure activities (spas and balneology) interacting with other cascade uses.

4.5. Oceania

In Oceania, the country that stands out in a matter of implementation and use of geothermal energy from direct uses to facilities for generating electricity is New Zealand. For this country, a geothermal system with multiple products is found at the Kawerau complex.

4.5.1. New Zeland

In the geothermal area of Kawerau, an Ormat binary cycle plant of 2.4 MWe plant was installed, afterwards, a 3.5 MWe plant from the same manufacturer was commissioned. These plants are operated altogether with a process plant of pulp and paper, which demonstrates the cascade utilization of geothermal steam and the direct utilization. Fig. 32 shows the scheme for this facility [78]. The temperature of the geothermal resource for this complex is 270 °C. In Fig. 33 shows a detailed scheme of the characteristics of the geothermal steam flow for the cascade stages in the Kawerau complex. It can be seen that the geothermal steam is intended for the production of electricity and drying wood products.

4.5.2. Region summary

For this region is reported the cascade utilization of high enthalpy geothermal resource. In this case it was possible to produce electricity not only in the first level of the cascade but also in an intermediate level interacting with pulp and paper processes at different thermal levels as well as drying of wood products. This system demonstrates that cascade use can be appropriate for industries with requires high amounts of steam and hot water and low electricity demand.

5. Cascade systems around the world

The literature review shows various schemes around the world where cascade utilization of geothermal energy is used. In particular, cascade systems that have been implemented practically, conceptual systems that have been studied from the point of technical and economic feasibility and research in progress trying to demonstrate the advantage of using geothermal energy in cascade are identified in order to maximize the rational use of the geothermal resources.

In summary and as one of the main contributions of this literature review Table 1 has been prepared, which includes the facilities and proposals of cascade utilization of geothermal energy chronologically arranged. It includes the country and the region, the reference from

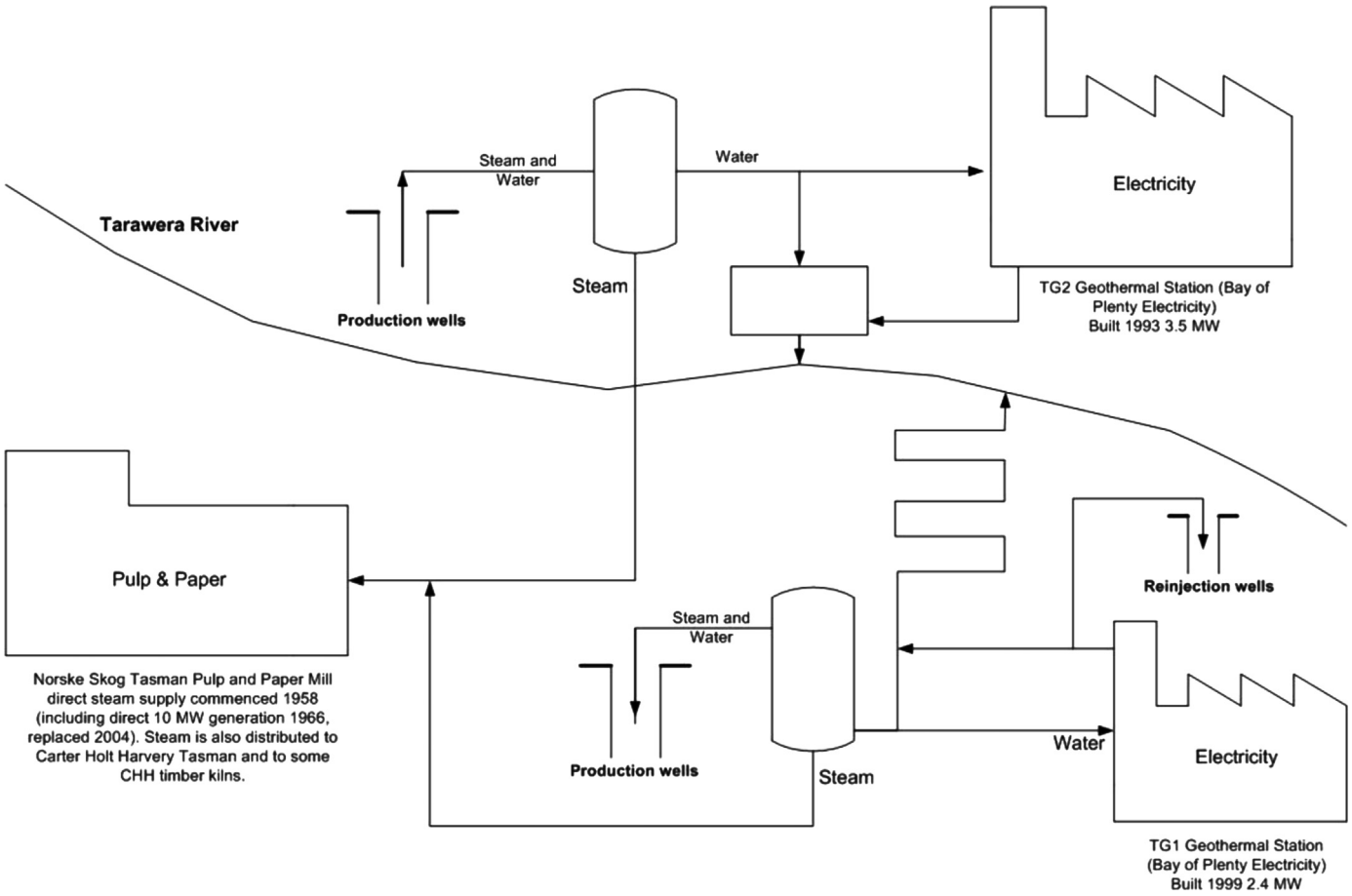


Fig. 32. Simplified design of the geothermal energy ratio for the Kawerau complex, outlined from [78].

which the information was obtained, the thermal levels of the cascade and the system products. For the year, the one indicated in the data relating to the facility or to the publication has been considered.

In general, there are two large groups of cascaded systems: A first group where geothermal heat is used directly to different temperature levels, and a second group in which there are systems that produce electricity at a first temperature level and in subsequent levels have other direct thermal uses. In both cases, various heat exchangers or thermally activated technologies are used.

As shown in Table 1, within the group of utilization of geothermal heat is: United States of America (Klamath Falls and Alaska), Poland, Austria (Geiberg), Greece, Macedonia, Kenya, Romania, Mexico and Hungary. In most of these countries geothermal energy for district heating and heat supply in greenhouses has been used. In some cases, the use of geothermal heat has been incorporated for drying processes, aquaculture, fish farming and balneology. For Alaska (USA), absorption machines activated by geothermal heat have been utilized to produce ice. In Koutso (Greece), both absorption machines and heat pumps have been utilized for district cooling.

With regard to the systems that provide electricity and heat for different applications, the corresponding to Iceland (Husavik), Oradea in Romania and Kawerau in New Zealand have stood out. The latter generates electricity and useful heat but is not a formal implementation in cascade of geothermal energy, but what can be considered as a multiple use of energy. In China it is also highlighted the compatibility of using geothermal resources for electricity generation through ORC or Kalina cycle integrated in oil processes for the production of crude oil.

There are other projects and research in progress that include cascade utilization of geothermal energy, as in the case of the project in progress at Canby, California. This project involves the production of electricity and the direct utilization of geothermal heat in district and greenhouses heating, and aquaculture. In this sense, the multicenter project for Kozani, in Albania, is another very interesting cascade utilization and covers electrical applications, the direct utilization of geothermal energy and even the incorporation of solar collectors.

6. Discussion and concluding remarks

From the review conducted, it has been possible to show that the concept of cascade utilization is an effective way to exploit in a sustainable manner the high potential of geothermal resources classified as medium and low enthalpy. Gradually the barriers that limited the utilization of this geothermal resources have been reduced due to recent technological development of low-temperature power cycles, such as binary cycles, organic Rankine cycles and Kalina Cycles. Such technologies are currently available in market to work properly with temperatures typically in the range of 80 – 120 °C that match perfectly with the range temperature of medium enthalpy resources and to form the first level of a cascade system. In the same way, thermally activated technologies as absorption and adsorption machines and heat pumps play an important role to enhance the benefits of cascade systems. These technologies usually require temperatures in the range of 50 – 90 °C making them good candidates to form the second level of a cascade system. On the other hand, there is a wide range of industrial, semi-industrial, agricultural and drying processes in the range of 30 – 70 °C that can be coupled sequentially or in parallel to form other levels of the cascade, bringing more benefits both economic and environmental. It is true that this type of systems requires high investments, but that is compensated with the various products obtained from a single renewable source or the combination with other conventional energy systems and other renewable resources. It is expected that information presented in this review provides enough theoretical and technical elements for understanding current cascade systems and implement new systems to other regions where the geothermal resources of medium and low enthalpy are available. The concluding remarks and specific comments of this review are presented below.

6.1. Cascade concept definition and benefits

Cascade utilization of geothermal energy is a reasonable sequential arrangement to use energy at different thermal levels and to obtain several products. The cascade implementation is a key proposal on how to optimize the rational use of energy, to

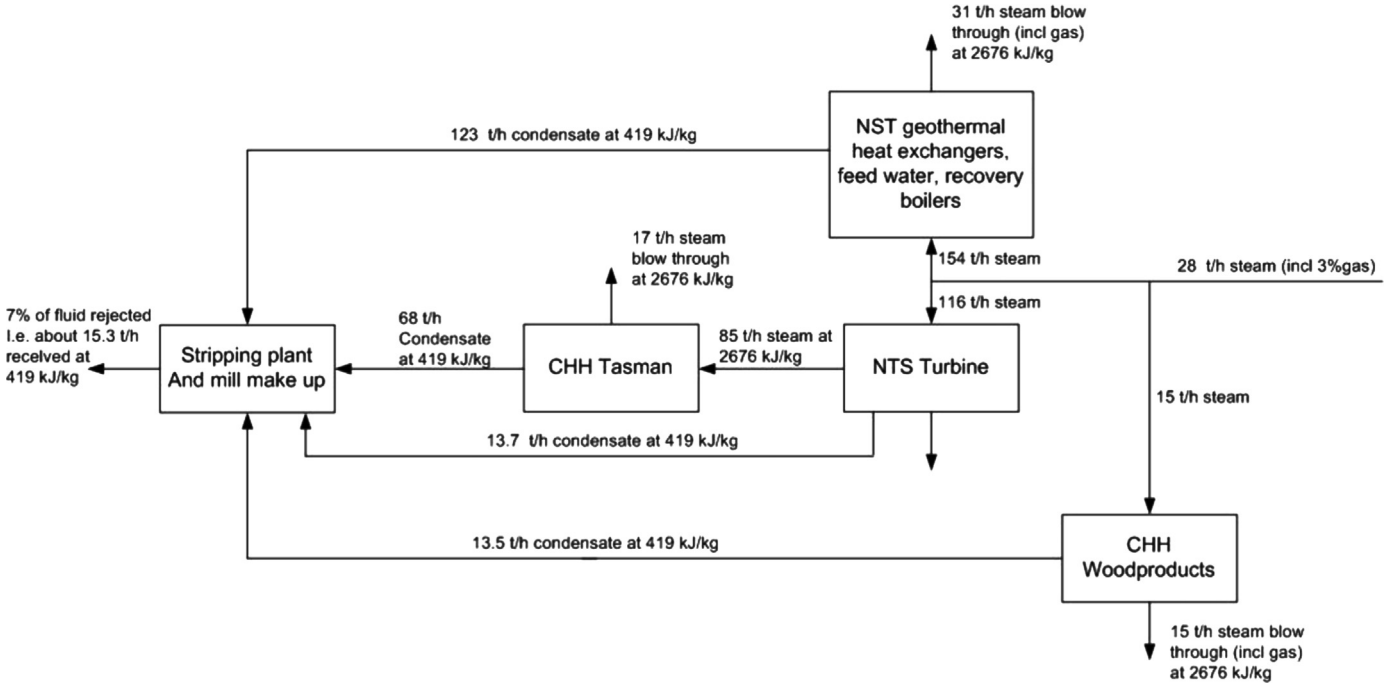


Fig. 33. Details of the cascade utilization within the Kawerau complex, outlined from [78].

Table 1
Facilities and proposals around the world with cascade utilization of geothermal energy.

Year	Country	Region	Ref.	Levels	Geothermal cascade product																		
					A	B	C	CB	CS	D	DH	DW	E	G	HW	I	IG	MS	O	P	PI	PP	SH	VG	WP
1974	Iceland	Sudurnes	[45,65]	2		X					X		X												
1987	Italy	Siena	[67]	2							X		X	X											
1989	Austria	Altheim	[45,55]	2		X					X		X												
1990	Iceland	South-Iceland	[45,66]	2							X		X												
1993	New Zealand	Kawerau	[78]	2									X												X
1994	USA	Klamath Falls	[23]	3										X			X			X					
1995	Germany	Neustadt-Gl.	[45,57]	2							X		X												
1995	Romania	Oradea	[72]	3		X					X		X	X	X										
1995	Romania	Oradea	[73]	3	X	X					X		X	X	X										
1995	Thailand	Egat	[42–45]	4					X	X			X				X						X		
1998	Iceland	Husavik	[62–64]	–		X					X		X	X		X		X		X					
1998	Poland	Podhale	[69]	4							X									X				X	X
1998	USA	Nevada	[22]	–						X			X												
1999	Austria	Geinberg	[56]	4		X					X			X									X		
2001	USA	New Mexico	[24,25]	2									X							X					
2002	Greece	Koutso	[58]	–			X		X					X								X			
2002	Greece	Langadas	[59]	–							X			X									X		
2005	Macedonia	Kocani	[68]	–												X									
2005	USA	Oregon	[21]	2									X							X					
2005	USA	Alaska	[26–28]	–		X	X							X											
2009	Poland	Szczecin	[70]	2									X												
2009	Slovenia	Lendova	[25,76]	–	X		X				X		X	X									X		
2010	Iran	Sabalan	[41]	–		X					X		X												
2010	Kenya	Eburru	[17]	4						X		X		X						X	X		X		
2010	Romania	Oradea	[74]	4							X				X			X							
2010	Turkey	Kutahya	[77]	4							X		X	X									X		
2011	Romania	Oradea	[71]	3		X								X								X			
2012	Canada	Ontario	[19]	3			X				X		X		X										
2012	China	Tianjin	[37]	2									X						X						
2012	Mexico	Maguarichi	[20]	4	X	X				X				X											
2013	Albania	Tirana	[47–54]	5	X	X					X		X	X	X					X			X		
2013	Hungary	Morahalom	[60,61]	–							X														
2013	USA	Canby	[29–31]	3	X						X		X	X											
2014	China	Shanghai	[40]	2			X						X												
2014	Kenya	Samburu	[18]	–				X		X			X				X			X					
2015	Romania	Oradea	[75]								X		X							X					

Description: A – Aquaculture, B – Balneology, C – Cooling, CB – Cattle Breeding, CS – Cold Storage, D – Drying, DH – District Heating, DW – Domestic Water, E – Electricity, G – Greenhouses, HW – Hot Water, I – Industry, IG – Irrigated Agriculture, MS – Melting Snow, O – Oilfield, P – Pisciculture, PP – Pasteurization Plant, SH – Spa and hotels, VG – Vegetables Growth, WP – Wood Products.

maximize the utilization of geothermal resources of medium and low enthalpy, to reduce the consumption of primary energy obtained from fossil fuels and to reduce the costs and the harmful gas emissions to the atmosphere.

Cascade utilization of geothermal resources allows obtaining multiple products from a single resource or coupling to other renewable sources, thus having a better utilization and energy saving, and avoiding an excess consumption of non-renewable resources. Cascade utilization corresponds to a particular case of integrated energy systems, also known as polygeneration systems, which focus on the principle of using one or more energy resources to provide various products more efficiently than the conventional systems can.

6.2. Characteristics of cascade systems

There are several possibilities to utilize geothermal energy in cascade manner, but in general, it is possible to identify two main groups: cascade systems with electricity production and thermal uses, and cascade systems comprising only thermal uses. Cascade systems with electricity production are suitable when geothermal resource has temperatures greater than 90 °C, with more opportunity of integrate various thermal levels as temperature increases. Cascade systems with only thermal uses are appropriate for geothermal resources with temperatures below 90 °C. However, this is not a limitation because it is possible to hybridize the system with other renewable sources, such solar energy, upgrading the geothermal heat until reach the requirement to produce electricity. All the cases described in this review make use of geothermal resource with temperatures ranging from 65.5 °C to 162 °C, with exception of Iceland and New Zealand that reported temperatures of 240 – 380 °C typical of high enthalpy resources. This fact confirms that cascade utilization is most appropriate to promote the sustainable use of low to medium enthalpy geothermal resource.

The main aspects of cascade systems around the world are presented as they were mentioned in the review:

- *Africa*: The temperature range suitable for cascading direct uses is 89 °C in wellhead and outlet temperature for the last level of the cascade of 50 °C. However, the cascade concept can be extended comprising electricity production, when the geothermal resource has 120 °C. Living standards of rural communities located near geothermal zones can be improved substantially by considering artisanal and industrial developments to form a new concept known as Geothermal Village. Climate conditions influence strongly some uses such as space heating, not reported for this continent.
- *America*: Most of the systems reported have included electricity production in the first level of the cascade, with exception of Mexico and Klamath Falls (Oregon, USA) that reported that shallow wells with temperature less than 98 °C are appropriate for agriculture and aquaculture direct uses in different thermal

levels. However, if temperatures range between 90 °C and 150 °C binary cycles or organic Rankine Cycles can be integrated increasing cascade levels. In this sense, old wells or exploration wells already drilled that were not suitable for power generation can be put into operation reducing capital cost. In some cases, the final uses or products in the cascade system are conditioned by the climate zones favorably. That is the case of Chena hot springs, where it was possible producing ice by using absorption machines and part of heat rejected was used as the second level for greenhouses. The performance of this system was increased substantially because of the cooling water for the absorber and condenser of the absorption machine was taken from a river whose temperature remains almost constant with 4 °C.

- *Asia*: The systems described for this region make use of medium enthalpy geothermal resource with temperatures as low as 83 °C, suitable for electricity production in the first level of the cascade. The temperature of geothermal fluid for cascade systems in Asia range between 83 °C and 140 °C. According to the studies corresponding to China, it is possible utilize oil wells as geothermal wells supplying heat for a cascade plant interacting with oil processes and replacing oil boilers. On the other hand, environmental temperature is a key factor in the performance of cascade plants, such in the case of Thailand where the power output in summer season was decreased up to 50% increasing heat rejected for the second level. The main advantages of cascading geothermal energy in all the cases described for Asia are both economic and environmental.
- *Europe*: Due to its geographic and climate characteristics Europe has undoubtedly the most wide variety of applications for cascade use of medium and low enthalpy geothermal energy. Main direct uses implemented in European cascade systems are heating networks, greenhouses and balneology, due to the particular needs of the continent. Currently, geothermal district heating has a share of about 58% of the total European cascade solutions. The countries that show greatest cascade use are Iceland, Austria, Greece and Romania. It is worth noting that the implementation of these technologies provides better quality use of energy in its counterpart with higher investment costs. Other feature for this continent is the advanced technological development and continuous research allowing the implementation of cascade systems for residential sector, industry, government and public services and agriculture. The living standards also allow focusing more on recreation and leisure activities (spas and balneology) interacting with other cascade uses.
- *Oceania*: For this region is reported the cascade utilization of high enthalpy geothermal resources. In this case it was possible to produce electricity not only in the first level of the cascade but also in an intermediate level interacting with pulp and paper processes at different thermal levels as well as drying of wood products. This system demonstrates that cascade use can be appropriate for industries with requires high amounts of steam and hot water and low electricity demand.

6.3. Integral utilization and hybrid systems

This cascade utilization has led to propose new concepts for the integral use of geothermal resources, as in the case of what has been called Geothermal Village, which aims to develop human settlements as close as possible to where geothermal resources are found. The Geothermal Village is presented as self-sustainable from an energy point of view, besides generating various economic activities around the use of geothermal energy, and promoting the economic and social development of communities. Another related concept is the creation of multi-center services including gym, spa pools and spas, including greenhouses and aquaculture activities. In these multicenters, it is possible to incorporate the use of other renewable sources, such as solar energy both thermal and photovoltaic.

6.4. Other benefits and some drawbacks

In general, it can be stated that the cases presented have a point of convergence of the benefits obtained from this form of utilization of geothermal resources. The main benefits that can be highlighted are: an increased profitability of the facility, maximized use of geothermal resources of medium and low enthalpy, local development of communities and cities, as well as social and environmental benefits. Furthermore, it is also possible to use various thermally activated technologies and processes, which can be tailored to the needs of each particular case. It is important mentioning that in some cases medium and low enthalpy potential has not been used for power generation due to high initial cost and limited technology availability, using only geothermal energy for direct-use applications. On the other hand, climate conditions influence strongly some geothermal uses such as space heating or district heating. Living standards are also an important factor because recreational uses are most appropriate for developed countries.

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